

Neutral Citation Number: [2025] EWHC 1764 (TCC)

Case No: HT-2023-000321

IN THE HIGH COURT OF JUSTICE
BUSINESS AND PROPERTY COURTS OF ENGLAND AND WALES
TECHNOLOGY AND CONSTRUCTION COURT (KBD)

Royal Courts of Justice
Rolls Building, Fetter Lane,
London EC4A 1NL

Date: 11 July 2025

Before :

MR JUSTICE CONSTABLE

- -

Between :

PHAROS OFFSHORE GROUP LIMITED

Claimant

- and -

KEYNVOR MORLIFT LIMITED

Defendant

- -
- -

Dermot Woolgar (instructed by Watson Farley & Williams LLP) for the Claimant
Thomas Macey-Dare KC (instructed by Holman Fenwick Willan LLP) for the Defendant

Hearing dates: 9-12 and 19 June 2025

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JUDGMENT

This judgment was handed down by the Judge remotely by circulation to the parties' representatives by email and release to The National Archives. The date and time for hand-down is deemed to be 10.30 on the 11th of July 2025.

Mr Justice Constable:

A. Introduction

1. Viking Link is a 1400MW high voltage direct current electricity link which transports power between the United Kingdom and Denmark, from one country to the other depending on supply and demand, connecting at Bicker Fen substation in Lincolnshire and Revsing substation in southern Jutland, Denmark. Viking Link is approximately 765 km long. It is said to be the longest land and subsea electricity cable in the world, and has the capacity to transport enough electricity for up to 2.5 million UK homes. This case concerns the burial or embedment of several kilometres of electrical and fibreoptic interconnector cable in the seabed immediately off a long stretch of sandy beach called Henne Strand or Houstrup Strand on the west coast of Jutland.

2. The Claimant, Pharos Offshore Group Limited ('Pharos') is a provider of subsea equipment, offshore personnel hire, and engineering consultancy services to the offshore oil, gas and renewable energy markets. The Defendant, Keynvor Morlift Limited ('KML') is a marine contractor. In circumstances more fully described below, Pharos provided the jet trenching equipment and personnel to operate the equipment to KML, which was itself engaged as a contractor to Prysmian Powerlink Srl ('Prysmian') to undertake various services in connection with a section of the bundled electricity cable which had been laid as part of the construction of Viking Link. Among the services which KML had been subcontracted by Prysmian to provide was carrying out the burial or embedment of the majority of this section of the cable, to a depth of 2m, together with various other associated services ("the Embedment Works"). Central to the equipment supplied by Pharos was a remotely operated underwater jet trenching vehicle, sometimes referred to as a TROV (a tracked remotely operated vehicle), referred to as the UTV-670.

3. Jet trenching involves the jetting of the seabed using seawater pumped at high pressure from the UTV-670, which moves along the route of, and astride, the cable, while jetting the seabed immediately next to the cable. Seawater is pumped onto and into the seabed through an array of nozzles along a pipe or "sword" attached to the undercarriage of the UTV-670. When deployed in use, the sword is extended down towards the seabed, rotating about its front or top end. The seawater which is pumped through the nozzles along the length of the underside of the sword "liquidises" the seabed to the depth to which the sword has been extended, and a further array of nozzles along the side of the sword propels seawater laterally towards the cable. A trench is thereby formed in the seabed, to the side of and beneath the cable. The cable settles under its own weight into the bottom of the trench. As the UTV-670 moves on, the seabed material is left to settle back into the trench on top of the cable.

4. There is a dispute about whether, in this case, there was a contractually binding commencement/mobilisation date. KML says that there was, and that it was 9 June 2022. There is no dispute that the UTV-670 and some associated equipment was refused permission for temporary export by the National Clearance Hub between 6 and 17 June. The equipment arrived on 20 June 2022. The works were not complete, through to and including demobilisation, until 17 August 2022. There were significant periods of downtime, due to weather and sea-state conditions, due to the difficulties encountered in particular in (but not limited to) the nearshore where slopes in excess of 15° were encountered, and due to various operational and technical

issues. The causes of the downtime, and legal responsibility for the delays attributable to those causes, lies at the heart of the dispute between the parties.

5. Pharos delivered 6 invoices to KML dated between 5 May 2022 and 31 August 2022 totalling £918,471.79 including VAT. There is no dispute that, of this, £135,048 has been paid. Pharos claims the balance, together with statutory interest. KML contends that the UTV-670 and associated services were not supplied in accordance with its specification and/or the agreement, causing loss and damage to KML. It contends that the maximum sum to which Pharos should have been entitled (but for KML's claimed set off and Counterclaim) was £274,742.05 (ex VAT). KML claims between £1,217,806.86 and £1,434,049.50 in respect of additional vessel/spread hire, harbour fees by way of damages and/or liquidated damages of £185,280.

B The Factual and Expert Witness Evidence

6. Pharos relied upon the factual witness evidence of Mr Phil Walker, the CEO and founder of Pharos, Mr Christopher Gamble, a Technical Director of Pharos and Mr Thomas Frazer, who was contracted by Prysmian as a consultant and 'Client Representative' during the Embedment Works.

7. All three gave evidence, Mr Frazer by video-link from Australia, and were subjected to cross-examination.

8. Each were honest in the manner in which they gave evidence before the Court. Whilst, in the circumstances described more fully below, Mr Walker admitted to prior dishonesty in some of his communications with the authorities whilst seeking to expedite the UTV-670's release from customs, I found no reason to doubt his evidence under oath which was generally straight forward and direct. Similarly, whilst it is likely that in one aspect of his written evidence he overstated the position with regard to the previous history of transportation of the UTV-670, in answers to questions under oath, Mr Gamble was candid, and fairly admitted difficulties and took responsibility for those difficulties appropriately. Mr Frazer's evidence was relatively limited and impressionistic given that he gave his statement two and a half years after the events in question, and did not refer to any contemporaneous documents. His oral evidence was straight forward and he answered questions directly. As set out in more detail below, an important aspect of his evidence relating to why the MBES survey was delayed was not 'demonstrably wrong', as asserted by Mr Macey-Dare KC on behalf of KML in Closing Submissions. His recollection was in fact, when judged against more probative documents than those deployed in cross-examination, demonstrably correct. Although there is, later in this judgment, one particular respect in which I do not accept his characterisation of events, I consider that Mr Frazer's recollection was broadly reliable, and uninfluenced by the fact that he had known (although had little social contact with) Mr Gamble for a long time. Mr Frazer was plainly trying, as best as he could, to assist the Court.

9. KML called Paul Scullion, Group Commercial and Financial Director of KML. Mr Scullion took time to think through his answers with care. I formed the view that his recollection seemed to fail at times when the question might otherwise suggest an answer unhelpful to KML's case. As I consider below in the context of whether an

express agreement preceded (and indeed 'warranted') the removal of certain words important to the question of the waiting on weather risk allocation, I do not consider his lack of recollection about the circumstances in which he came to sign a statement of truth to have been genuine. My overall view was that Mr Scullion's priority was to help KML's case rather than the Court.

10. KML also relied upon the witness statement of David Freeburn, a freelance contractor, but working under contact as a Site Marine Manager for KML. His evidence was critical of Pharos's equipment and personnel. Mr Freeburn was unavailable for the trial and his evidence was tendered under a Hearsay Notice. The circumstances of Mr Freeburn's unavailability were vague. Where it is now routine to give live evidence from pretty much wherever a witness may be located and in whichever time zone (as demonstrated by Mr Frazer, giving evidence – undoubtedly at much inconvenience to himself - from Australia where he was attending a family member's wedding), I place little weight on the untested evidence of Mr Freeburn.

11. Both parties called expert evidence to deal with the technical issues and analysis of the causes of downtime during the Embedment Works. Pharos called Darren Moore, Principal Engineer at BPP Technical Services Limited. KML called Alex Scott, principal of his own marine consultancy. As Mr Moore candidly accepted, Mr Scott had more practical experience than Mr Moore. In their oral evidence, both witnesses understood their duties to the Court and gave answers seeking to assist the Court in understanding how they saw things. As explained in more detail below, Mr Moore's detailed analysis of the causes of downtime was more granular and particularised, and therefore reliable, than the equivalent exercise by Mr Scott. However, rather than concluding that the approach of one expert was to be preferred over the other in all respects, I have accepted or rejected various aspects of both experts' opinions, for the reasons I give below.

C. Factual Chronology up to Sailaway of the Susanne A

12. On 21 December 2021, Phil Walker and Paul Scullion, on behalf of Pharos and KML respectively, entered into a Mutual Confidentiality Agreement in respect of Pharos' potential involvement in providing project-related support to KML in respect of the Embedment Works. On the same day, Pharos supplied the UTV-670 Data sheet and a 'Track Record' document. The UTV-670's specification included '*Trenching speeds*' of '*200m/hr in <40kPa*', and '*Pitch & Roll*' of '*15° side slopes 15° fore and aft slopes*'. The covering email stated: '*This looks to be ideal against your requirements'*.

13. In response to this, KML asked for '*a schedule of rates*', and Pharos replied, '*Please see attached indicative pricing*'. There is a 'rental' proposal dated the same day to supply KML with a UTV-670 Trenching spread for all or part of the associated requirements for such operations on a permanent, or standby basis, inclusive of equipment, personnel and management requirements, stating that Pharos understood the required tasks, and was experienced in such, for these types of operations. Section 6 dealt with 'Commercial' and was split into 'Equipment Lump Sum Fees' and Daily Rates as follows, in the sums which eventually became part of the agreement between the parties:

6.2 Equipment Lump Sum Fees

No	Description	Price (GBP)	Basis
1	Project Management and Initial project documentation package (pre-job planning, risk assessment's, UTV670 procedures)	24,000	Lump Sum
2	Subsequent documentation / engineering requests	Cost + 15%	Lump Sum / Event
3	Mobilisation/Demobilisation	Cost +15%	Lump Sum / Event

6.3 Daily Rates

No	Description	Price (GBP)	Basis
1	UTV670 Rental Rate (Equipment only)	4,250	Daily Rate
2	Personnel (as defined in section 5.1.1) Based on 7 Man Team	4,550	Daily Rate

14. On 24 December 2021, Mr Scullion confirmed that Prysmian had issued to KML a Letter of Intent covering the engineering and documentation elements of the project.

15. Discussions continued, and on 9 January 2022 KML provided Pharos with a collection of maps and charts from Prysmian which included two "*Alignment Sheet Charts*". Neither of these charts indicated the presence of slopes greater than approximately 4°.

16. On 10 January 2022, a Letter of Intent was provided from KML to Pharos, again limited to the preparation of documentation.

17. On 18 January 2022, Mr Shaw of Pharos sent three documents to Mr Scullion and others of KML, namely a "*Shallow Water Trenching Method Statement*", a "*Risk Assessment & Method Statement*" and a "*Shore End Storyboard*". As to these:

(1) The "*Shallow Water Trenching Method Statement*" stated that the UTV-670 "*has the ability to drive and manoeuvre in and around inclines of -+15°*". It noted that the seabed conditions were "*Sand/Silt, patches of sand. Sand waves 2° gradient*". In Section 5, entitled, '*Recommended Trencher*', it was stated:

> '*Upon review of the available data to date, we believe the topography and soil conditions are favourable for Jet Trenching operations with the use of Pharos Offshores' UTV-670 Jet trencher….. The selected TROV has the ability to drive and manoeuvre in and around inclines of -+15º, ensuring safe and ease of movement around the cable route.'*

(2) The "*indicative trenching performance*" on the available information was given as 250m/hr, with two passes likely to be required to achieve "*a target trench depth of 2.0m*". Under '*Trenching Profile*', part of Section 7, the Method Statement, the document stated that:

> '*[a]chievable trench depth and trenching system rates of progress may be affected by local seabed features including slopes and uneven seabed, which are always possible…From the information provided by the client we do not anticipate any major slopes over 2 degrees*'.

(3) In Section 7.5, under Trench depth control, the document warned "*that there is a risk that local conditions can be encountered which are outside the specified operating conditions of the vehicle, or outside the control of the operating personnel…. Pre engineering is key to the success of any project, a full review of the soil's information supplied by the client along with the required specification will be conducted prior to the start of offshore operations*".

(4) The "*Shore End Storyboard*", showed that Pharos was, at this point, proceeding on the basis that the proposed support vessel was a barge which would "*come up the beach on high tide*" and anchor on spuds.

18. On 28 January 2022, Mr Scullion informed Mr Taylor and others that KML was "*now in receipt of a letter of award for the project*". On 4 February 2022, Mr Scullion then circulated a programme for comments on this date which proposed 5 days for trenching, as part of a total programme lasting 62 days (albeit one which appeared to have a significant gap between the equipment transiting to site and onsite commissioning). There is no documentary evidence suggesting that anyone at Pharos suggested that the programme was wholly unrealistic.

19. On 10 February 2022, KML issued its first draft (designated "*Rev B01*") of a Project Method Statement for Prysmian. A specification sheet for the barge *Mormaen* was appended. A few days later, on 15 February, Mr Scullion indicated that Prysmian was considering postponing the cable burial scope '*to give less weather exposure*'. A further email on 24 February indicated that KML wanted a discussion to see whether the Embedment Works would work around '*a window from say 22 March to maybe early April…This would mean that we could work around any weather delay in pull in and your availability…*'

20. Prysmian provided its purchase order to KML for the execution of the landfall works on the Danish side in the frame of the Viking Link project on 7 March 2022. At this stage, the Embedment Works formed an option, which was later exercised.

21. On 8 March 2022, emailing Mr Scullion the previous chain of emails in which the 'Rental Proposal' had been provided, Mr Taylor said, '*As discussed we can offer a standby/weather rate of £3k per day for equipment only*'. An excel spreadsheet, seemingly produced by Mr Scullion, dated the same day includes a breakdown of the various costs related to the Embedment Works. The excel shows the rates provided by Pharos (e.g. £4250 for the equipment and the personnel per day), with whatever mark-up KML was making (e.g. 25% or 30%). The analysis included a section within which KML priced 'weather', assessed at 10 days, at the £3,000 rate for equipment provided within this email (and retaining, as per the email, £4,250 for personnel when waiting on weather). It showed 6 days for the cable burial works and 2 days mobilisation and demobilisation.

22. The method of calculating 6 days is itself shown (in lines 1 and 2 and columns R to W in the 'Separate Burial' sheet) in the spreadsheet referred to above. The basis of the calculation was explained by Mr Scullion in his evidence, not challenged on this point, which I broadly accept insofar as it identifies the basis upon which KML arrived at 6 days based upon the indicative data provided by Pharos:

> "*In the specifications provided by Mr. Shaw (Pharos) on 21 December 2021, Pharos indicated that the Tool had a trenching speed (i.e. a burial rate) of "up to 300m/hr (120 m/h in <50 kPa based on previous performance)". Subsequently, Pharos indicated in the Initial Draft Method Statement that the Tool had a burial rate of 250m/hr (e.g. 4.2 m/min). In any event, KML, in attempting to be prudent, wanted to allow for contingencies in its own internal cost modelling and project planning (e.g. KML wanted to calculate the expected timeframe on a conservative basis, to allow for some contingency time). Accordingly, KML determined that, even if KML allowed for a burial rate of 0.5 m/min (i.e. only 10% of that indicated in the initial specifications, or about 12% of that subsequently indicated) and assumed that the Tool could only operate for 60% of a 24-hour day (e.g. just over 14 hours, instead of 24-hour operations), it would take the Tool up to 6 days to complete the cable burial scope of 2,360m (as shown in the screenshot below). Together with 2 days for mobilisation, 1 day for calibration and 1 day for demobilisation, this was the basis upon which it was agreed that KML would pay Pharos for 10 unweathered days for hire of the Tool, equipment and Pharos' personnel*."

23. I note that the spreadsheet does not accord precisely with this evidence: it appears to allow 2 days for mobilisation and demobilisation (rather than 3) and nothing for calibration. There is no documentary evidence that this spreadsheet or the calculation specific to the 6 days was shared with Pharos.

24. By 20 March 2022, a subsequent draft of the Project Method Statement had been reviewed by Prysmian. This document indicated that KML was proposing, instead of a barge, to use a "*DP [Dynamic Positioning] multicat*" instead.

25. Prysmian sought an assurance from KML that the proposed TROV could work with a partially-buried cable "considering the time spent from the time of pull in operation will be 2 months" and was told "Yes, pre burial survey with a TSS350 will confirm the depth and position of the cable prior to trenching" [E 79 3]. Prysmian asked about the expected speed of trenching and was told "*speed of 150m/hour is estimated by Pharos for this scope*". Prysmian asked specifically about the "side slope limit" of the proposed TROV, stating "*That must be clear in case of ROV shall across steep a sand dune [sic]*" and was told that, while the vehicle had "*proved exceptionally manoeuvrable to date, the OEM supplied documentation suggest a maximum of ±15°/ 4m for both pitch & roll*". By reference to an internal Pharos review document, it was put to Mr Gamble that on the basis of a previous project, experience had shown that, '*it has proved itself very limited at manoeuvring with a charged feed hose connected'*. A charged feed hose is connected whilst the UTV-670 is actively trenching. In light of this, it was suggested that Pharos' answer was misleading. Mr Gamble denied this, responding that the limitations related to circumstances in which the UTV-670 was offset from the vessel. At the point of answering this question, the limitations of the Suzanne A were not known. However, it is clear that there were at least some circumstances, which ultimately became relevant, in which the type of indicative speeds provided were not going to prove possible.

26. On 4 April 2022 KML issued revision B03 of its Project Method Statement in which it identified the proposed support vessel as the *Coastal Challenger*, "*a DP1 multicat vessel with shallow draft*". On the same date, KML indicated by email that "*We need to trench in shallow water (too shallow for a vessel) for approx. 200m, this would*

either be done with the TROV spread on a barge moored as close into shore as possible or with a trailer on the shoreline".

27. Amongst a series of exchanges between Jake Rowland Crosby for KML and Mr Shaw of Pharos, Mr Rowland Crosby emailed Mr Shaw on 8 April identifying that '*we have just had the slope angle at site through from our surveyor…The nearshore sandbank in the east (right) of the photo is where we see the sharpest gradient of about 13deg. Possibly there will be a sharper gradient in the nearshore section as well, we are surveying this early next week. Do you have any thoughts on this with the TROV in mind? Will it be OK with these gradients?'*

28. On a similar note, the same day, Mr Scullion asked, amongst other things:

> *"We believe that the sandwaves are likely to give a depth variation of c.2.5m on the seabed. Robin, our surveyor, is confirming the pitch of these waves and we will share this information with you as soon as it is available. Please could you confirm that your unit can manage this incline and trench successfully to the depth that we require?"*

29. In response, Mr Shaw asked if the seabed "*is stable or is it constantly moving*" and asked about the distance between "*the megger ripples*"; he then asked whether thought had been given by KML to "*running our mass flow down the corridor before we drive the trencher down the route*", suggesting that if this was done shortly before trenching "*it would certainly reduce the angles and reduce risks*".

30. Various discussions continued about the topography in the very nearshore and its potential implications. On 29 April 2022, KML issued version B04 of its Project Method Statement. This described, for the first time, the use of a guide wire, or winch wire, to support the hose and umbilical when the TROV was operating in the nearshore and very nearshore area, to KP0.23. There was evidently a meeting on this date between the parties at which this proposal was discussed.

31. The first purchase order (No. 16229) between the parties was dated 5 May 2022. This related to preliminary work related to a CAD Storyboard and a series of meetings. The total value of that Purchase Order was £9,750 inch VAT. The subsequent invoice relating to this purchase order was paid. On the same date, KML provided a deck layout for the *Susanne A*, the vessel KML eventually used, the choice of which features in a number of the allegations and counter-allegations made by the parties.

32. On 10 May 2022, KML issued version B06 of its Project Method Statement, in which the support vessel was identified as the *Susanne A*. This version also stated that an MBES would be carried "*Before TROV operations…To confirm seabed conditions… Confirm lay of the sand banks, any inclinations of concern*". The document also included the following by way of Weather Limits:

8.2 WEATHER LIMITS

Weather limits will be governed by the offshore activities of the trenching operations with the vessel safety of the SUSANNE-A operating in shallowing water being the critical aspect.

Prior to an operation commencing the provided weather forecasts will be analysed to see if a suitable weather window is available and agreed by all parties.

Operation	In charge	Duration [h]	Contingency [h]	Wind [m/s]	Wave Hs [m]	Visibility in Air [m]	Current (m/s)
Workboat Operations	KML	4	2	10	1	500	N/A
TROV Deployment / Recovery	KML	2	1	10	1	500	0.5
Trenching – Shoreline to ~KP0.25	KML	24	12	10	1	500	0.5
Trenching – KP0.25 – KP2.36	KML	48	24	10	1	500	0.5

Table 8-1 - Weather limits.

NOTE: The Vessel Masters will be responsible for the safety of their own vessels and will have the final say on weather conditions.

33. Mr Moore, Pharos' expert, understood that the durations of the suitable weather windows from the trenching rows provided a duration plus contingency for the relevant works (albeit by reference to the final version which allowed 24+12 hours for inshore trenching and 48+24 for offshore trenching). Mr Scott, when cross-examined, also held the view that these were intended to indicate the period of time required to carry out the relevant works. I accept this agreed evidence as what can be objectively understood by the technical document. It makes sense. The parties obviously anticipated undertaking each section of trenching in one go if that were possible, as that would be the most efficient way of carrying out the work, and these were the durations anticipated by reference to the distance taking into account a conservatively 'discounted' trenching speed. I note, however, that this version suggests an aggregate period of 4.5 days (and, in the later version, 5 days) rather than the 6 days calculated by Mr Scullion.

34. The next purchase order (No. 16249) was provided dated 10 May 2022 by KML, under cover of an email dated 16 May 2022 ('the 10 May Purchase Order'). There is no dispute that the 10 May Purchase Order incorporated the standard terms and conditions of KML (set out insofar as relevant in Section D below). The main part of the purchase order was as follows:

Description	Unit	Unit cost	£
Project Management and documentation	1	24,000	24,000.00
UTV670 and supporting tooling and equipment per day	10	4,250	42,500.00
Personnel per day	10	4,550	45,500.00
Plus weather downtime at applicable rates			

Scope of Work

- Onshore project management, pre-job planning, risk assessments, provision of procedures and operational plans, pre- and post-job reporting as required
- Mobilisation and demobilization of equipment spread and personnel. Equipment on hire from arrival at port of mobilisation. Transportation payable at cost + 5% on presentation of invoices. Personnel mobilisation payable at 1 day each way.
- Burial of cable bundle (2xHVDC,1xFO) to minimum 2m DOB along cable route to KP2.36 in accordance with specifications provided by KML and indicative programme provided by POG.
- Basis Susanne-A vessel platform as agreed with POG team, 24 hour operations during available weather windows.
- Confirmation of DOB of cable using TSS350 system (noting quotation for survey equipment to be agreed).
- Transportation and permitting for personnel and equipment including completion of RUT registration and customs clearance as required.

Total	**112,000.00**

VAT AS REQUIRED TO BE CHARGED AT THE APPLICABLE RATE
IN ACCORDANCE WITH KML STANDARD TERMS & CONDITIONS

35. On 17 May 2022, further to the fifth bullet point in the 10 May Purchase Order which had noted that the quotation for survey equipment was yet to be agreed, Jeff Taylor of Pharos provided a list of the required survey equipment.

36. The Action Tracker on 20 May 2022 indicated that Pharos was to review the 'BAS' (a Burial Assessment Study) to confirm the suitability of the TROV. Paul Scullion was also tasked with issuing a revised PO. The Tracker was emailed to the Pharos and the KML team under cover of an email of the same date which referred Chris Gamble of Pharos to the '*BAS I sent you early today, please can you review*'.

37. The BAS had been prepared for the main contractor in February 2022 by a company called Asso Subsea. On considering the part of this BAS which related to the Embedment Works, Pharos understood that the seabed might consist not just of sand and silt, but that it might also contain gravel or shale or some other dense granular material. These heavier materials tend to accumulate in the bottom of the trench before the cable settles into the trench, and can prevent the cable from being buried to the required depth. To address this, Pharos considered it may be necessary to "overtrench" (i.e. create a deeper trench than would ordinarily be necessary), and arranged to bring a 1m extension piece to Denmark, to extend the reach of the 2.1m sword by 1m.

38. In a WhatsApp exchange on 23 May 2022 between Jeff Taylor and Paul Scullion, Mr Taylor asked for the payment terms of 60 days to be reduced to 30 days, as to which Mr Scullion said that he would see what he could do.

39. On the same day, Mr Taylor emailed Paul Scullion of KML stating:

> '*As discussed we have some extra procurement to be done for the project which needs to be done by CoB tomorrow at latest to meet project timelines as well as start extra fabrication for the trencher for survey equipment to be fitted.*
>
> *Can you confirm we can invoice against original PO for these works please and can you send an amended PO to cover all the extra works please?'*

40. In a second email a couple of hours later, Mr Taylor said,

> '*Please see table below for estimated extra costings. These are all worst case scenarios so there should be some savings overall but if we can get the PO to cover this that would be perfect.'*

41. The table then set out various equipment, including lay flat hoses and the sword extension, stating that the '*Total Inc of Cost+10%*' to be £81,103.

42. There is no suggestion in the documentation that there was any other communication between the parties about any other aspects of the 10 May Purchase Order.

43. On 24 May 2022, KML issued revision B07 of its Project Method Statement in which it provided details of the means by which the umbilical and hose would be attached to the winch wire.

44. The next day, Mr Scullion wrote to Mr Taylor attaching a revised purchase order, and re-attaching the terms and conditions:

> '*Further to our discussions and your email below, please find attached an uplifted PO to reflect the additional items discussed.'*

45. The equivalent section of the replacement purchase order, dated 23 May 2022, was as follows ('the Revised Purchase Order'):

Description	Unit	Unit cost	£
Project Management and documentation	1	24,000	24,000.00
UTV670 and supporting tooling and equipment per day	10	4,250	42,500.00
Personnel per day	10	4,550	45,500.00
Additional equipment per Jeff Taylor email 23/05/22 15:36 to include oroflex hose, fracking set, swivel joints, flanges, rigging, sword extension and deployment arm. Value is NTE and will be rechargeable at cost +10% on production of supporting invoices	1	81,000	81,000.00

Scope of Work

- Onshore project management, pre-job planning, risk assessments, provision of procedures and operational plans, pre- and post-job reporting as required
- Mobilisation and demobilization of equipment spread and personnel. Equipment on hire from arrival at port of mobilisation. Transportation payable at cost + 5% on presentation of invoices. Personnel mobilisation payable at 1 day each way.
- Burial of cable bundle (2xHVDC,1xFO) to minimum 2m DOB along cable route to KP2.36 in accordance with specifications provided by KML and indicative programme provided by POG.
- Basis Susanne-A vessel platform as agreed with POG team, 24 hour operations during available weather windows.
- Confirmation of DOB of cable using TSS350 system.
- Transportation and permitting for personnel and equipment including completion of RUT registration and customs clearance as required.

Total	**193,000.00**

VAT AS REQUIRED TO BE CHARGED AT THE APPLICABLE RATE

IN ACCORDANCE WITH KML STANDARD TERMS & CONDITIONS

46. It can be seen that additional wording has been introduced to reflect the 'cost+10%' additional equipment, with the sum of £81,000 stated and 'value NTE'. This means Not To Exceed. This additional wording is located where the words, '*Plus weather at applicable rates*' had previously been, which words no longer appear. The scope of works has been reduced in font size, and amended to remove the parenthesis in the fifth bullet point.

47. Mr Taylor responded thanking Mr Scullion for the Revised Purchase Order, but asking '*As discussed, can we have some amendments to payment terms agreed?*', setting out a justification for the request. There is no dispute that this was, then, agreed.

48. Both parties accepted that there was a variation effected by the Revised Purchase Order. The as varied agreement constituted the agreement between the parties ('the Contract').

49. KML's "Action Tracker" created on 26 May 2022 noted "*Works schedule – Mobilisation date: 08/06/22*". The following day, Mr Gamble of Pharos informed Mr Thomas of KML that he still needed a shipping address as "*presently booking for departure UK 6th June, ETA 9th AM Esbjerg*".

50. On 27 May, Mr Gamble provided information to Mr Thomas and Mr Rowland Crosby in relation to the water hose couplings which evidently had been discussed earlier between them. The Senior Project Engineer for KML, Mr Berry, responded, '*Jake has just forwarded me this e-mail. This solution looks good. Do you have any idea of the tensile strength at each joint?*'. Mr Gamble replied:

> "*Unfortunately I don't have anything on the tensile joint of these couplings, which are a bit of a late entry into the equation due to supply issues with our*

> *favoured option. The lay flat-hose to which they attach is rated to 37,000kg, internally we were hoping to get a load test performed, but we are still awaiting delivery and suspect there will be insufficient time pre-project to complete this scope.*"

51. Mr Berry then answered, on 30 May, indicating that he understood and that he would think they were fairly robust due to their purpose but considered it would be good to check if possible. He continued, '*I have only seen these connections used on land/deck - Is the plan to lock the release handle somehow for offshore use, perhaps cable tie it in position?*' Mr Gamble said, in response to this, that they had been advised that the latching handle could be bolted once in the closed position.

52. The "Action Tracker" created on 31 May 2022 noted "*Works schedule – Mobilisation date: 09/06/22*".

53. On 1 June 2022, Pharos issued a letter addressed "*To Whom It May Concern*", seeking assistance with the temporary export of the UTV-670 trenching system, explaining that the system would be undertaking shallow water burial works as part of the Viking Link project. The letter noted that the project "*is scheduled to commence mobilisation onto the salvage vessel Susanne A…on the 9th June 2022 in the port of Esbjerg*", and that the works were "*estimated at 14 days (may extend to 30 days to allow for local weather constraints)*". The UK customs' position is evidenced on three of the consignment notes, dated 6 June 2022. These were issued for the export of the UTV-670 and the other equipment. Reflecting the difficulties that were then encountered, they are marked "*Permission to progress not granted to this entry*".

54. Between 7 and 17 June 2022, both parties made repeated and extensive efforts to accelerate customs clearance for the UTV-670 and related equipment. It is clear that the National Clearance Hub were being chased daily for updates. The email exchanges also suggest that there was no particular issue with the paperwork which had been submitted. Both Mr Walker and Mr Scullion were active in writing to various governmental departments in trying to accelerate the release of the UTV-670. In one email exchange on 15 June 2022, Mr Walker wrote to the Department for International Trade, and suggested that Pharos faced going bankrupt as a result of the implications of the delays. Mr Walker accepted, in cross-examination, that this was a lie. In another email, on 17 June 2022, he suggested that Pharos' client was going to terminate the contract, '*as they/we will not be able to absorb the vessel currently on hire for ten days further*'. Mr Walker accept that this, too, was not true. Although the same line of cross-examination was not taken with Mr Scullion of KML, I note that in his email of 15 June 2022, Mr Scullion indicated that the vessel spread on hire was not working as a result of the absence of the trenching equipment, whereas even on KML's own case no delay had actually been caused by the non-arrival of the equipment by 15 June because of the weather conditions. This too, therefore, was somewhat of an overstatement of the true impact of the customs delays, no doubt with the intention of seeking to put pressure on the authorities. This Court plainly does not condone any of the communications in which the true impact of the absence of the UTV-670 was misrepresented to the authorities. However, I do not draw the conclusion from Mr Walker's admission in evidence that he had lied in these emails that he was not candid in his evidence before me.

55. On 17 June 2022, confirmation was received that the goods were cleared to proceed.

56. Meanwhile, on 10 June 2022, KML issued revision B10 of its Project Method Statement. This noted that "*the sand waves and beach are very mobile and likely the bank arrangement, slopes and positions will have changed since the last MBES. The MBES survey prior to operations [7] will confirm final positions of the sand waves and hence the positions the vessel will take to complete the operations*".

57. On this date, the consecutive daily sequence of detailed progress reports prepared contemporaneously by the parties started. Pharos produced an "*Equipment Daily Report*" and KML produced a "*Daily Progress Report*" every day up to and including 18 August 2022 (and KML produced one more "*Daily Progress Report*" on 19 August 2022). A series of daily calls between Prysmian, KML and Pharos also started, which were followed up by an email reporting the contents of the calls. I refer to these in more detail below.

58. On 20 June 2022, the three delayed consignments arrived at the Port of Esbjerg. The two pumps had already been mobilised onto the deck of the Susanne A and the remaining items, including the UTV-670, were then mobilised on the vessel over the next two days. On 22 June 2022, the Susanne A sailed from Esbjerg for Henne Strande, and works commenced in the late evening of 23 June 2022.

D. The Contract

59. As set out above, there is no dispute that the 10 May Purchase Order and the Revised Purchase Order were both subject to KML's terms and conditions.

60. For the purposes of the terms and conditions, KML was the 'Buyer' and Pharos the 'Seller'. They included the following:

> *1. DEFINITIONS*
>
> *In these Conditions, unless the context otherwise requires, the following words have the following meanings:*
>
> *"Contract" the contract for the purchase of Goods or Services constituted by the Seller's acceptance of the Purchase Order in accordance with these Conditions;*
>
> *2. GENERAL*
>
> *2.1 These Conditions, and any agreed formally authorised amendments thereto, shall govern the Contract to the exclusion of any terms and conditions which the Seller may seek to impose.*
>
> *2.2 The Purchase Order constitutes an offer by the Buyer to purchase the Goods subject to these Conditions and supersedes all prior discussions and negotiations.*
>
> *3. VARIATION*

3.1 No variation of these Conditions or of the Contract shall be effective unless agreed in writing between an authorised representative of the Seller and an authorised representative of the Buyer.

3.2 The Buyer shall be entitled to vary a Purchase Order and any specification that may relate to such Purchase Order, at any time prior to delivery. In this event, the Seller reserves the right to re-negotiate the terms of the Purchase Order including the price and delivery timescale.

4. QUALITY AND DESCRIPTION

4.1 The Seller warrants to the Buyer that the Goods or Services shall:

a) be of satisfactory quality and free from defects in design, material or workmanship and be fit for their intended purpose;

c) be of sound material and workmanship;

7. PAYMENT

The Buyer shall pay for Goods and Services within 60 days of delivery provided that they are in accordance with any applicable Specification and the terms of the Contract. Without prejudice to any other right or remedy, the Buyer reserves the right to set off any sum of money owing at any time by the Seller to the Buyer against any sum payable by the Buyer to the Seller under the Contract or under any other current contract between the Buyer and the Seller.

8. DELIVERY, DELAY, LIQUIDATED DAMAGES AND OTHER CONSEQUENCES

8.2 The Seller shall deliver the Goods or Services by the date specified in the Purchase Order. ...

8.3 Should the Seller fail to deliver the Goods or Services by the specified date for any reason other than Force Majeure, the Buyer (without prejudice to any other right or remedy which it may have) shall be entitled to, whether as a deduction from any sums due to the Seller or otherwise, 10% of the total Contract Price due to the Seller in respect of those delayed Goods or Services for each week (or part thereof) in which the Seller fails to deliver them.

8.4 Should the Seller fail to deliver the Goods or Services by the specified date for any reason other than Force Majeure, the Buyer (without prejudice to any other right or remedy which it may have) shall be entitled to:

c) charge to the Seller any additional costs, losses or expenses which the Buyer may reasonably incur due to the Seller's failure to deliver the Goods or Services by the specified date.

9. RISK AND DAMAGE/LOSS IN TRANSIT

9.1 The Goods or Services shall remain at the Seller's risk until delivered and accepted by the Buyer.

10. TIME

Time for performance and all other dates specified in the Purchase Order shall be of the essence. The supply of Goods and Services by the Seller to the Buyer may be time or project critical, and thereby a critical path item. The supply of Goods or Services by the Seller to the Buyer may be crucial for the timely and efficient operation of ships and vessels or marine facilities. The Seller must not delay or impede the operation of the Buyer's ships and vessels or marine facilities.

12. REJECTION

12.1 If any of the Goods or Services do not comply with the Contract or the Seller does not comply with its obligations under clause 4, the Buyer shall be entitled to reject those Goods or Services or any part of them by notice in writing to the Seller given within 14 days or such other period as may be agreed …

13. INDEMNITY

13.1 The Seller shall indemnify the Buyer against all liability, claims, costs and expenses (including reasonable legal expenses), loss, personal injury or damage to property which the Buyer may suffer or incur as a result of or in connection with:

a) any act or omission (including negligence) of the Seller, its sub-sellers, employees or agents in supplying, delivering and installing the Goods or Services in accordance with the Contract and the Conditions.

b) a breach of any warranty given by the Seller in relation to the Goods or Services including, but not limited to, the warranties contained in clauses 4 and 15;

14. ASSIGNMENT AND SUB-CONTRACTING

14.2 The Seller shall not sub-contract the Contract or any part of it other than for materials, minor details or for any part of the Goods or Services of which the makers are named in the Purchase Order or Specification, without the consent in writing of the Buyer. Any such consent shall not relieve the Seller of any of its obligations under the Contract.

15. SERVICES

15.1 Where the Seller is providing any services, the "Goods or Services" as used in these Conditions shall be interpreted to include the provision of such services. In providing those services, the Seller warrants that it shall use only qualified and experienced personnel acting with all care and diligence and it accepts full responsibility for such personnel, their property, their acts and omissions.

> *18. SELLER TO INFORM ITSELF*
>
> *The Seller shall be deemed to have satisfied itself as to the extent and nature of the Goods and Services; in the case of Services including but not limited to the services, personnel, materials and equipment, plant, consumables and facilities required … general/ local conditions, access, climatic, sea, water and weather conditions, and all other matters which could affect progress or performance of the Seller's obligations.*

61. It was alleged by KML, and admitted by Pharos, that the Contract was subject to an implied term, pursuant to section 13 of the Supply of Goods and Services Act 1982, that the works required of Pharos would be carried out with reasonable care and skill and, (if contrary to KML's primary case, there was no fixed duration) within a reasonable time. Pharos denies that it was an implied term that mobilisation would be carried out within a reasonable time, although it accepts an obligation to do what it reasonably could to get its equipment to the embedment works site.

62. There are a number of issues between the parties as to the correct construction of the Contract.

The Proper Approach to Construction

63. The principles applicable to the exercise of construction of the Contract are well established. A helpful summary of approach is contained in Popplewell J's judgment in Lukoil Asia Pacific Pte Ltd v Ocean Tankers (Pte) Ltd [2018] 1 Lloyd's Rep 654 at [8]:

> "*The court's task is to ascertain the objective meaning of the language which the parties have chosen in which to express their agreement. The court must consider the language used and ascertain what a reasonable person, that is a person who has all the background knowledge which would reasonably have been available to the parties in the situation in which they were at the time of the contract, would have understood the parties to have meant. The court must consider the contract as a whole and, depending on the nature, formality and quality of drafting of the contract, give more or less weight to elements of the wider context in reaching its view as to the objective meaning of the language used. If there are two possible constructions, the court is entitled to prefer the construction which is consistent with business common sense and to reject the other. Interpretation is a unitary exercise; in striking a balance between the indications given by the language and the implications of the competing constructions, the court must consider the quality of drafting of the clause and it must also be alive to the possibility that one side may have agreed to something which with hindsight did not serve his interest; similarly, the court must not lose sight of the possibility that a provision may be a negotiated compromise or that the negotiators were not able to agree more precise terms. This unitary exercise involves an iterative process by which each suggested interpretation is checked against the provisions of the contract and its commercial consequences are investigated. It does not matter whether the more detailed analysis commences with the factual background and the implications of rival constructions or a close examination of the relevant language in the contract, so long as the court balances the indications given by each."*

Daily Charges for more than 10 days

64. KML's case is that, on the true construction of the contract, Pharos was not entitled to claim more than 10 days of daily charges in total. Mr Macey-Dare KC contends that this is the effect of:

(1) Clause 2.2 of KML's standard terms, which provides that its purchase orders constitute offers on the terms contained therein which supersede all prior discussions and negotiations;

(2) Clause 6, which provides that the Contract Price is "*fixed and firm*" and cannot be increased without KML's prior written consent;

(3) Clause 3.1, the No Oral Modification provision;

(4) The 10 May Purchase Order, which provides for daily charges in a fixed amount per day for 10 'units', which go to make up the stated "Total" of £112,000 plus VAT;

(5) The Revised Purchase Order, which leaves those daily charges unchanged.

65. Pharos disputes that there was a cap or limit on the amount of per diem charges. It says that if the Embedment Works were properly to take longer than 10 working days, Pharos would be entitled to charge accordingly. It accepts that Pharos was not entitled to take as long as it liked: it was under an obligation to carry out and complete the Embedment Works using the UTV-670, the supporting tooling and equipment, and its personnel, with reasonable care and skill, and within a reasonable period of time. Mr Woolgar argues that Clause 6 does not assist KML as the 'Contract Price' is not £112,000. He says that on a proper construction '10' is not a 'not to exceed' number of days. The 10 days is '*indicative'*, to use the language of the third bullet point which refers to the '*indicative programme*'. Mr Macey-Dare KC contends that this is too vague a reference and probably refers to the sequence of works which by then had been included within the Method Statement.

66. At a forensic level, Mr Woolgar makes the point that this argument appears to have been something of an afterthought, because it was not an argument deployed in the Defence. Rather than contending that the '10' within the PO constituted a fixed period, KML's case at paragraph 11(2) of the Amended Defence is that there was a later amendment by a variation in writing as set out in the trackers of bi-weekly meetings that:

> '*The trenching work was to take 10 days. This was calculated using a progress rate of 50m/hr (which was significantly lower than the 250-300m/hr which the Claimant represented was possible, allowing 3 days mobilisation and 1 day demobilisation plus an additional margin [of] about 1.3 days.'*

67. I consider that, objectively, the number '10' within the Contract was an indicative number representing the then understanding of the parties of the likely duration of the works. It was not a 'not to exceed' or fixed number of days within which Pharos were contractually obligated to complete the works. This is the objective meaning for the following reasons:

(1) The Revised Purchase Order, like its predecessor, explicitly refers to the works being carried out pursuant to an '*indicative programme*'. This is an important phrase in the document and cannot be ignored.

(2) Neither party pointed specifically to a specific document which constituted the '*indicative programme*'. However, on the basis of:

(a) The stated '*indicative*' trenching speeds from Pharos' proposal.

(b) The Rental Proposal, which also included '*indicative*' pricing.

(c) The draft programme which was exchanged (albeit early on) showing 5 days anticipated trenching time, excluding mobilisation and demobilisation.

(d) The durations contained in the agreed Method Statement.

(e) KML's (albeit internal) spreadsheet which was based upon Pharos' '*indicative*' trenching speeds.

the sensible, objective interpretation of the Contract, read as a whole and in the factual context known to both parties at the time, is that the parties' reference to '*indicative programme*' conveyed a shared understanding that the Embedment Works were anticipated to take around 10 days (weather exclusive) when carried out with reasonable skill and care, but that this duration was '*indicative*' and not a fixed and guaranteed programme.

(3) Put another way, the express obligation to carry out the works in accordance with a programme said to be '*indicative*' is inconsistent with an objective understanding that the number of days is contractually fixed.

(4) 'Programme' ordinarily means more than a mere sequence of works, as contended for by Mr Macey-Dare KC. The sequence of works was not 'indicative' but fixed within the Method Statement which, by reference to 'KML's specification' also in the third bullet point, Pharos was obliged to follow. What was plainly agreed to be '*indicative*' was the period of time then anticipated by the parties in which the sequence of works would be completed. It is clear on the evidence that at various times and in various documents the parties had put their mind to the anticipated duration.

(5) The factual context against which the Purchase Order was issued is wholly consistent with an objective understanding that the anticipated duration of works was not fixed at 10 days. In particular, the parties were aware that (a) the actual conditions encountered in the seabed would affect the performance of the UTV-670 and (b) the actual conditions to be encountered were to be the subject of an MBES survey of the cable route post contract, but before the works themselves began; and (c) were in any event 'mobile' and could change even during the course of the works. It is unsurprising that, the programme was agreed by the parties (in a document drafted by KML) to be no more than '*indicative*'.

(6) Mr Walker's (unchallenged) evidence was '*…when we provide indicative costings, they are provided on the understanding that the costings are indicative and cannot be taken as a guarantee of the work taking a particular amount of*

time. This is a shared understanding between the parties. When Pharos provided indicative costings to KML, the situation was no different. We clearly stated when providing the costs that they were indicative. Paul Scullion, who as explained above was our main point of contact at KML, has had many years in the industry and I expected him to understand this'. Although Mr Scullion's evidence was that the period of 10 days was understood by him (subjectively) to be on a 'not to exceed' basis, he appeared to accept the common understanding referred to by Mr Walker in the following exchange:

> 10 Q. Yes. So I am suggesting to you that the phrase
> 11 "indicative programme" indicated that ––whatever
> 12 you might have hoped, I mean, no doubt reasonably
> 13 you wanted this work to be done as quickly as
> 14 possible , but you could have no guarantee that it
> 15 could be achieved within any particular timescale?
> 16 A. Yes, I think that is a fair statement. Equally, I
> 17 think it is fair to assume that where we are
> 18 provided with metrics regarding the performance of a
> 19 tool, that we would expect the performance to be
> 20 somewhere within the reasonable bands that we are
> 21 given.
> 22 Q. That depends upon the seabed conditions?
> 23 A. Agreed.

(7) If the reference to '10' units meant a fixed 10 day duration, the parties would, in effect have been agreeing a fixed lump sum for the entirety of the works at £112,000 (subject to weather). This is not what the PO says. Instead, it expressly distinguishes between one part of the contract price which is fixed irrespective of duration (Project Management and documentation) and other parts which are calculated per day, namely equipment and personnel (rates). KML's construction ignores this clear distinction.

(8) Furthermore, the distinction between lump sum and (uncapped) daily rates mirrors Pharos' Rental Proposal of 21 December 2021, which also referred explicitly to the trenching speeds and costings being 'indicative'. This is a document which, whilst not forming part of the contract, forms part of the context against which the objective understanding would be drawn. Indeed, Mr Macey-Dare KC relies upon this document to inform how the personnel rate was arrived at in the context of the 7-personnel issue, considered further below. It cannot be relied upon in one context and ignored in another.

(9) Where the parties intended 'not to exceed', they used this language expressly: see the agreement with respect to the cap of £81,000 in respect of additional equipment costs.

(10) 'Contract Price' is defined as the 'price or prices set out in the Purchase Order'. This is consistent with the potential for more than one price within a purchase order being a 'Contract Price' for the purposes of Clause 6. On a proper construction of the Contract, what is 'fixed and firm' for the purposes of Clause 6 is the lump sum price, for Project Management and documentation, and the 'Unit

> Costs' of the UTV-670 and supporting tooling and equipment per day, and of personnel per day. The number of 'units' required ultimately to complete the work was not fixed, but instead reflected the then '*indicative programme*', as understood by the parties.

68. Therefore, Pharos' entitlement to daily charges for equipment and personnel is not capped at 10 days. The works were required to be carried out in a reasonable time, and pursuant to the 'indicative programme' of 10 days from arrival at the port of mobilisation.

69. A reasonable time is to be assessed by means of a broad consideration, with the benefit of hindsight, of what would, in all the circumstances, have been a reasonable time for performance, taking into account matters such as any estimate given of how long it would take Pharos to perform (here, specifically, the 'indicative programme' and a period of 10 days), whether that estimate has been exceeded and, if so, in what circumstances; the causes of delay; and the extent to which they were within Pharos' control. As stated by Lord Herschell in Hick v Raymond & Reid [1893] AC 22 at page 29:

> "*I would observe, in the first place, that there is of course no such thing as a reasonable time in the abstract. It must always depend upon the circumstances. Upon "the ordinary circumstances" say the learned counsel for the appellant. But what may without impropriety be termed the ordinary circumstances differ in particular ports at different times of the year. As regards the practicability of discharging a vessel they may differ in summer and winter. Again, weather increasing the difficulty of, though not preventing, the discharge of a vessel may continue for so long a period that it may justly be termed extraordinary. Could it be contended that in so far as it lasted beyond the ordinary period the delay caused by it was to be excluded in determining whether the cargo had been discharged within a reasonable time? It appears to be that the appellant's contention would involve constant difficulty and dispute, and that the only sound principle is that the "reasonable time" should depend on the circumstances which actually exist."*

70. A reasonable time would therefore take into account the extent to which the works could reasonably proceed on account of the weather, including wind and wave height. It could be said that, if this is so, the inclusion of the words '*Plus weather downtime at applicable rates*' is surplusage and that the words would only add to the contract if the 10 days was itself a fixed and firm period. Mr Macey-Dare KC fairly accepted that the presumption against surplusage is relatively weak in commercial contracts. As noted at [7.24] of *Lewison, The Interpretation of* Contracts (8th Edn), Patten LJ observed in Al-Hasawi v Nottingham Forest Football Club Ltd [2019] EWCA Civ 2242 that "*arguments based on surplusage or redundancy are rarely reliable or sure ground on issues of construction*". Mr Macey-Dare KC relies (through paragraph [7.27] of *Lewison*) on the observation of Coulson J (as he then was) in Secretary of State for Defence v Turner Estate Solutions Ltd [2015] EWHC 1150 (TCC) that although a presumption against surplusage was of little weight in interpreting a standard form of contract, it has a part to play in the case of "*a bespoke contract carefully drafted by the parties to meet the exigencies of this particular and significant commercial arrangement*".

71. In the present case, the Revised Purchase Order, and its predecessor, were of course bespoke documents. To describe them as '*carefully drafted by the parties*' would, however, be something of an overstatement. They contain a brief description of what was to be required using relatively loose language – they do not, for example, identify what the 'specifications provided by KML' were, or, as already discussed, identify with any precision the 'indicative programme' other than by reference to 10 per diem units. That the document would carry with it the presumption of meticulous legal drafting seeking no redundancy is somewhat unrealistic. The argument based on surplusage is insufficiently weighty to justify the conclusion that '10' units is to be construed as imposing a fixed and firm duration, rather than a reflection of the explicitly referred to '*indicative programme*'.

Daily Charges for more than 7 personnel per day

72. KML contends that the Contract provides for payment of 10 days of daily personnel charges at a unit cost of £4,550 per day (plus VAT). It is said that this represents £650 per person for 7 persons. Mr Macey-Dare KC argued in Closing Submissions that the commitment was limited to 7 persons per date at £650 per person per day, with a total daily personnel charge of £4,550 (plus VAT).

73. Mr Woolgar states that this is not pleaded. Mr Macey-Dare KC accepted that Mr Woolgar was correct that this point was not a point raised before by KML. It is, indeed, right that the Amended Defence does not explicitly refer in narrative to any apparent issue relating to the number of personnel permitted to be charged for each day. However, Mr Macey-Dare KC contends that this is a point of construction and raising it causes no prejudice.

74. The difficulty with taking the point is that KML's case, at a detailed level, on the quantification of Pharos' claim both as opened and, indeed, closed is not predicated on Pharos being limited to 7 personnel.

75. At commencement of trial, I was provided with a spreadsheet said to be the fruits of the joint labour of both sets of solicitors, although principally produced, I was informed by Mr Woolgar, by KML's instructing solicitors. I shall refer to this as 'the Master Spreadsheet'.

76. The Master Spreadsheet included two sheets containing, '*D's position on C's Inv (Daily)'* (Tab 5) and '*C's Position C's Inv (Daily)* (Tab 7). Within Tab 5, KML's position on the appropriate daily personnel charge recoverable by Pharos reflects the actual number of personnel deployed, rather than a number capped at 7. In a document produced for Closing, referred to in more detail below as KML's Counterfactual, KML again quantifies its case on charges owing to Pharos on various assumed counterfactuals, on the basis of the actual number of personnel deployed, taken from Tab 5 of the Master Schedule.

77. Given the clarity of KML's acceptance of Pharos's claim for the actual number of personnel in its own quantum calculations, presented both in opening and closing, it is not necessary to consider the contractual point raised for the first time by Mr Macey-Dare KC in his written closing submissions. Even if I were wrong about this, I would not consider it appropriate to allow an amendment to the pleadings to permit KML to argue that, on a proper construction of the Contract, Pharos' entitlement was to deploy

and be paid for no more than 7 personnel. It was a point on which Pharos would have been entitled to call evidence in respect of, how and why 'additional' personnel were deployed, and whether that was a position acceded to by KML at the time. It is prejudicial for the point to be taken explicitly only at the end of the trial.

Personnel mobilisation and expenses for more than 7 personnel

78. The same analysis applies to this point. Whilst a number of points are taken with respect to expenses within Tabs 5 and 6 of the Master Spreadsheet presented in opening, and incorporated into the £55,584.41 admitted (on the assumed basis of full liability for the duration of the works) by way of expenses at cell 78/AF of the Counterfactual, no point has been taken that the expenses should be limited to 7 personnel.

Daily charges for personnel prior to the arrival of the UTV-670

79. Pharos deployed Mr Gamble and Neil Thirlwall to Esbjerg on 8 June 2022 in anticipation of the scheduled arrival of the UTV-670 on 9 June. For the reasons explained above, the UTV-670 did not in fact arrive until 20 June. In Closing Submissions, Mr Macey-Dare KC accepted that Pharos was entitled to deploy Mr Gamble and Mr Thirlwall to Esbjerg a day or so prior to the arrival of the UTV-670, and to charge daily rates for those personnel on those dates, pursuant to the provisions of the Contract which defined the scope of work as including "*pre-job planning*" and "*Mobilisation and demobilisation of equipment spread and personnel. Equipment on hire from arrival at port of mobilisation*". He contends that the Contract cannot, however, sensibly be construed as entitling Pharos to do so 12 days before the arrival of the UTV-670.

80. KML is correct: there is no proper contractual basis for charging for the deployment of these people from 8 June 2022 onwards when the Contract explicitly provided that equipment was on hire from its arrival at the port.

Daily charges during periods of breakdown

81. In Closing, Mr Woolgar fairly accepted that there was no contractual basis upon which Pharos could charge for equipment or personnel during periods of delay due to breakdown, where those delays were due to Pharos' breach of contract, even where the delay was concurrent with waiting on weather. This would follow from KML's obligation under Clause 7 to pay for Goods and Services '*provided that they are in accordance with any applicable Specification and the terms of the Contract*'.

Daily charges for waiting on weather

82. KML's case is that the effect of the issue of the Revised Purchase Order, in which there is no dispute that the words '*Plus weather downtime at applicable rates*' were removed, is that the risk of waiting on weather transferred from KML to Pharos. In relation to KML's plea that the Contract was amended by the removal of the relevant words, Pharos asked in a Request for Further Information whether it was KML's case that, before the creation of the Revised Purchase Order, one of the parties had

proposed to the other that the Contract should be varied so as to remove Pharos's right to payment during periods of weather downtime (question 1) and/or whether the parties had agreed this so that the Revised Purchase Order was intended to give effect to this agreed variation (question 3). KML's response to questions 1 and 3 was as follows:

> "*The parties were aware that the original time frame for completion and the price set out in the Purchase Order contained risk allowances associated with weather. This can be seen from table 8.2 of document PPL17017-IE-S-058 dated 10 May which preceded the Purchase Order, and which was co-authored by the Claimant. The Purchase Order was revised to reflect this, i.e. the reference to weather downtime was deleted because an allowance for weather down time had already been taken into account. The 10 days provided for in the Purchase Order included 3 days for mobilisation, 1 day for demobilisation and 4 days operational/ working time (which based on table 8.2 was 78 hours/ 20 hour working day) thereby allowing for 2 days of weather risk. Thus, there was no proposal by one party to the other, rather an acknowledgement weather down time had already been priced in and therefore the term relating to weather downtime was to be taken out. The parties agreed verbally that the variation to the purchase order was warranted.*
>
> *This will be a matter for witness evidence in due course.*"

83. In answer to a further question, KML stated specifically that the removed words '*were not merely overwritten*'. In other words, KML's positive factual case was that, in contrast to any suggestion that the removal of the words was the result of an unintentional over-writing of the text, the removal was a deliberate one, seeking to reflect a verbal agreement reached in respect of reversing the previously stated risk of waiting on weather.

84. A statement of truth accompanying this answer was signed by Mr Scullion.

85. In fact, no evidence supporting KML's pleaded positive case was served. The author of the Revised Purchase Order - someone unidentified from KML - was not called to give evidence. Mr Scullion's evidence at paragraph 35 of his first witness statement was limited to the following:

> "*Further to telephone conversations with Mr. Taylor (Pharos), on 25 May 2022, I circulated [07/102] a revised version of the Purchase Order which incorporated the additional items agreed (including, for example, an extension to the jetting sword to allow the Tool to reach the required cable burial depth of 2m) [02/49]. Mr. Taylor acknowledged receipt of the Purchase Order and provided no feedback on the Purchase Order save to request a variation of the terms of payments from 60-days to 30-days, in light of the fact that the cable burial works were only expected to last 10 days [23/369]. Save for the payment terms, Pharos raised no issue with – or otherwise indicated that they did not agree to – the variations set out in the Purchase Order circulated on 25 May 2022*."

86. In cross-examination, Mr Scullion confirmed that he had not had any conversation with anyone at Pharos about the transfer of risk, and could not think of anyone else at

KML who would have had such a conversation. When asked why, if he could not substantiate any discussion of the sort set out in the Response, he had put his name to the statement of truth, Mr Scullion simply said that it had been signed over a year ago, and he could not now recollect the conversation referred to.

87. Having heard that evidence and considered the documentary record, I conclude that it is not remotely likely that there was in fact any conversation between anyone in KML and Pharos in which they discussed and agreed verbally that a change to the Purchase Order '*was warranted*' in order to reverse the previously agreed risk allocation in relation to waiting on weather. It is inconceivable that if such an important point had been discussed in the context of revising the Purchase Order an explicit reference to that discussion would not have found its way into the email exchanges at the time. Moreover, I accept Mr Walker's unchallenged evidence that he never agreed to such an amendment, and that if KML had proposed such an amendment through Mr Taylor, Mr Taylor would have sought his approval before agreeing. I also accept Mr Walker's belief that this did not happen. Regrettably, I am unable to accept that when he signed the Statement of Truth in relation to the Further Information in May 2024, there would have been any reliable basis for Mr Scullion to have thought that such a verbal agreement had preceded the production of the Revised Purchase Order. If there had been a proper basis for thinking the Further Information was true, that is not something that Mr Scullion would since have simply forgotten.

88. Given that I have rejected KML's explanation that the removal of the words was purposeful, in giving effect to a verbally agreed position, the most likely remaining explanation for the removal of the wording is simply that the new text unintentionally over-wrote the words 'Plus weather downtime at applicable rates'. This is so irrespective of the fact that there no doubt could have been room to retain the wording had the author of the document thought carefully about it. As is tolerably clear from the wording of the email accompanying the revised Purchase Order, the sole purpose of providing the Revised Purchase Order, construed objectively, was to uplift the previous Purchase Order to reflect the additional equipment for a not to exceed sum of £81,000 following Jeff Taylor's email of 23 May 2022 at 15.36. It is no doubt in light of this obvious fact that KML felt it necessary to support its case by asserting that the parties had in fact put their minds purposefully to removing the phrase to reflect a new common understanding which had evolved since the 10 May Purchase Order. However, that assertion is, as I have found, wholly without foundation. There is no proper basis to conclude from the Revised Purchase Order, set in its factual context including the circumstances in which the words came to be removed, that the offer made by KML as constituted within the Revised Purchase Order was based upon a fundamentally different allocation of risk in relation to waiting on weather to that which had been agreed previously.

89. Whilst therefore Clause 2.2 makes clear that the Purchase Order constitutes an offer which supersedes all prior discussions and negotiations, this does not mean that as a matter of contractual construction, that offer must be construed in a complete vacuum from the factual context in which the offer was made. An argument based upon Clause 2.2 is circular, in that it is necessary first to construe what, objectively, the 'offer' was. Mr Macey-Dare KC is, of course, correct that there is nothing within the revised PO itself, viewed in splendid isolation, to indicate that the omission of the operative words was "*anything other than deliberate*". However, it is not appropriate

to look, objectively, only to the revised PO itself, in isolation from the factual context in which it sits. When sat in its wider context, it is plain that the reasonable person in the position of the parties would have concluded that the objective intention of the Revised Purchase Order was to insert the entitlement to additional payment for the additional equipment, leaving the previously agreed risk allocations unaffected, notwithstanding the unintentional loss of some of the words.

Application of LADs

90. KML claim liquidated damages between the date upon which KML contends (at least in its pleading) that Pharos ought contractually to have concluded the Embedment Works and the date it in fact did so.

91. It does so on the basis of Clauses 8.2 and 8.3. Clause 8.2 provides that '*The Seller shall deliver the Goods or Services by the date specified in the Purchase Order*'. Clause 8.3 provides that, '*Should the Seller fail to deliver the goods or Services by the specified date for any reason other than Force Majeure, the Buyer (without prejudice to any other right or remedy which it may have) shall be entitled to…10% of the total Contract Price…for each week (or part thereof) in which the Seller fails to deliver them*.'

92. No date was specified in the Revised Purchase Order, or its predecessor. Moreover, as I have found, the obligation as to time was one to complete in a reasonable time in accordance with the indicative programme, rather than a fixed and firm duration of 10 days. Mr Macey-Dare KC correctly accepted that there would be no general right to liquidated damages if the obligation was to complete within a reasonable time.

93. Whilst it was canvassed during exploration of the issue in Closing Submissions that, providing there was a fixed mobilisation date, liquidated damages would run until delivery of the UTV-670 ready for mobilisation, this cannot be right. The liquidated damages provisions affix to delivery of the services later than the date specified in the contract. KML's case takes this to be the date upon which the Services were to be completed. KML's pleading does not allege that the mobilisation date it says was agreed as 9 June 2022 was 'the date specified' for the purposes of Clauses 8.2 and Clauses 8.3. Even if it had been, such a claim would fail. The date for delivering the equipment to port – in effect at the very start of the services to be delivered under the Contract - could not sensibly be considered to be 'the specified date', as though it were some sort of milestone within a sophisticated machinery attaching liquidated damages to such milestones. That is simply not the way Clause 8.3 worked.

94. In the circumstances, KML had no right to liquidated damages for Pharos's delays, whether in the context of the arrival of equipment at Esbjerg at the start of the Embedment Works or their completion.

The Mobilisation Date

95. Irrespective of the question of liquidated damages, Mr Macey-Dare KC argues that the parties expressly agreed that Pharos would deliver the TROV and associated equipment at Esbjerg on 9 June 2022 such that a failure to achieve this placed Pharos in breach of contract. As set out above, KML accepts that 9 June 2022 was not specified anywhere in the Purchase Order(s). Mr Macey-Dare also made clear that the agreement in respect of the delivery date did not operate by way of any variation to the Contract (perhaps on the basis that such an argument would fall foul of the no oral variation provision at Clause 3.1).

96. It is said, however, that the agreement of the date constituted the operation by the parties of the contractual mechanism for fixing the delivery date, which they must be taken to have agreed in the contract. Mr Macey-Dare KC accepts that this alleged contractual mechanism was also not explicit. Instead, it is argued that the parties 'must have contemplated' that they would agree a delivery date, once KML's schedule for the works and the available weather windows became clearer.

97. Although not articulated as such, Mr Macey-Dare KC is effectively arguing that there is an implied contractual mechanism for fixing the delivery date, pursuant to which the date was agreed and became binding.

98. However, such an argument does not meet the test for the implication of terms. Whilst it may be obvious as a matter of practicality that the parties would want to liaise about co-ordinating the delivery and mobilisation dates, it is neither obvious nor necessary (in circumstances where the parties did not in fact include anything of the sort in their written agreement) that the results of such co-ordination were intended to become contractually binding obligations. It is trite that the process of implying a term must not become the re-writing of the contract, and that as such the term must be formulated with precision. Yet no part of the asserted implied 'contractual mechanism for fixing the delivery date' is expressed with precision. What is the implied mechanism? Would the agreement of the date need to be in writing? In a particular document? Would unilateral insertion of a date into a tracker be sufficient to fulfil the 'contractual mechanism' and bind the other party? What was the nature of the unexpressed obligation to meet the mobilisation date? It is neither obvious nor necessary that, had the parties put their minds to it explicitly, they would have agreed an absolute obligation, in circumstances where Pharos would be reliant upon third parties and matters generally outside its direct control (e.g. the customs process). Might the obligation agreed have been one of reasonable endeavours? Or some other expression of obligation, short of an absolute one? There would be a range of different ways of expressing the obligation, each with its own commercial common sense depending on the viewpoint, and it would be not right for the Court to select one and impose it as being the 'obvious' or 'necessary' one.

99. Mr Macey-Dare KC expressed the view that, if left to the 'statutory' implied term about time for performance, it was unsatisfactory because the obligation to mobilise 'within a reasonable time' might permit Pharos to deliver the UTV-670 ready for mobilisation at a time which, although 'reasonable' was inconvenient for KML in terms of the mobilisation of its own vessel, which itself might not be ready for the UTV-670 and give rise to standing charges. This may be so, and it is for reasons such as this no doubt that the parties did in fact co-operate over the mobilisation date. But it is not an observation which itself assists in the identification of a single, obvious and/or necessary, articulation of how the parties would have come to express the

contractual machinery for fixing the delivery date and the parties' obligations arising therefrom for the purposes of implying a term. It is just not possible to conclude what the parties 'must have contemplated' in this respect.

100. In circumstances where the date of 9 June 2022 was not agreed pursuant to an implied contractual machinery (as alleged), or itself as a variation to the Contract (not alleged), the only obligation upon Pharos was to carry out the overall works, of which deliver of the UTV-670 was a part, within a reasonable time. It had to carry out its obligation to export the UTV-670 with reasonable skill and care. This is a different obligation to one to ensure that the UTV-670 was delivered within a reasonable time. However, in case I am wrong about this, I consider not only whether Pharos breached its obligation, but also KML's alternative case that there was a discrete temporal obligation to deliver the UTV-670 within a reasonable time, in the following section.

D. Did Pharos deliver to the UTV-670 to Esbjerg with reasonable skill and care and/or within a reasonable time for the purposes of mobilisation?

101. KML contends that a reasonable time for mobilisation of the UTV-670 and associated equipment was by 9 June 2022. It relies on the fact that this was the date scheduled and agreed (if not in a binding contractual sense) by the parties, and on the fact that the cause of the delay beyond that date was sufficiently within Pharos' control. KML refers to its action tracker dated 31 May 2022 (item 109) which Pharos accepts (in contrast to other trackers produced during the project) that it received. I accept that this was the scheduled date agreed between the parties (albeit not in a binding sense). It is consistent with, for example, Mr Walker's letter 'To Whom It May Concern' in respect of the temporary export of the UTV-670 on 1 June 2022, which stated that, '*The project is scheduled to commence mobilisation onto the salvage vessel Susanne A…on the 9th June 2022*'. It is also supported by the fact that KML arranged to mobilise the SUSANNE A at Esbjerg on 10 June 2022 and Pharos dispatched the UTV-670 and equipment from Blyth on 6 June 2022 with an anticipated delivery date (at least in accordance with the transit time of 3 days, excluding any delays in customs) of 9 June 2022. Pharos also mobilised Mr Gamble and Mr Thirlwell to Esbjerg on 8 June 2022.

102. Another aspect of what a reasonable time is, seen in hindsight and taking into account all the circumstances as they transpired, is when the UTV-670 was actually needed. Put another way, even if delayed beyond its scheduled delivery date, delivery would still be within 'a reasonable time' providing that it did not cause actual delay or disruption to KML's other activities. Although the parties have addressed this question in the context of causation, it is equally relevant to the question of what a reasonable time is for delivery.

103. Mr Scullion asserted in his witness evidence that as a result of the delays to the UTV-670 in customs, KML missed the good weather windows which were available from 9 June 2022. He contends that if the UTV-670had arrived on 9 June 2022, and operated as it should have done, the cable burial works would have been completed with almost no weather downtime by 26 June 2022. The witness evidence from Mr Gamble, for Pharos, echoed by Mr Frazer, was that, when the UTV-670arrived on site on 23 June 2022, the trenching works could not commence immediately anyway because the method statement required KML to first complete the MBES survey works before the Embedment Works could commence, and these had not been

completed by 23 June 2022 because of bad weather. Specifically, although the MBES survey had already been completed in respect of the offshore and nearshore sections of the cable route, it had yet to be completed in the very nearshore section as the weather conditions had not permitted this. The MBES was completed the day after the date the UTV-670arrived. In response to Mr Gamble's statement, Mr Scullion's written evidence was that the reason the MBES Survey was carried out as close to the commencement of the trenching works as possible was because of the potential mobility of the seabed, and not because it could not be done sooner because of weather conditions. The 'late' MBES Survey was, therefore, directly linked to the delayed delivery of the UTV-670.

104. These competing contentions were explored in evidence. Having considered that evidence together with the relevant documents (and, in particular, the notes from the Daily Calls which were circulated each day from 10 June 2022 onwards by Matteo Borselli of Prysmian, copying in numerous relevant personnel including Mr Scullion, Mr Gamble and Mr Frazer, whose respective evidence I have referred to above) I reach the following findings of fact:

1) The MBES survey was '*stopped due to the weather*' on 9 June 2022. At this point, it was not anticipated that it would be able to resume shortly: the next workable window to complete the MBES survey was 15 June 2022. The suggestion by Mr Scullion in his written evidence that, had the UTV-670 arrived on 9 June 2022, it could have been immediately deployed is simply wrong. Although Mr Scullion's initial oral evidence appeared to be that there was a conscious decision that the MBES was only partially carried out because of a conscious decision to ensure that the very nearshore area was surveyed as close as possible to the deployment of the UTV-670, he ultimately accepted in cross-examination that the reason for stopping on 9 June 2022 would have been the weather. There is no contemporaneous evidence to support the existence of any instruction to carry out a partial survey, and it is clear from the Daily Call notes that the actual reason for stopping the MBES survey was because of the weather.

2) According to the weather limits proscribed in the Method Statement for the operation of RHIB 2 (the vessel that would be undertaking the nearshore MBES), the wave height limit was 0.5m, subject to the express caveat that the Vessel Masters would be responsible for the safety of their own vessels and would have the final say on weather conditions. As accepted by Mr Moore in cross-examination, and as is apparent from the agreed data, from 13.00 on 15 June 2022 to 13.00 on 17 June, there was a period of 48 hours during which the weather conditions did not breach the limits for the carrying out of an MBES survey. However, it might be noted, of the 48 hours, all but 2 hours were either at the limit (0.5m) or just under (0.4m). Neither expert appears to have considered the influence of tides on the ability to carry out the very near shore MBES survey during this window.

3) On Tuesday 14 June 2022, the Daily Call notes record in respect of the MBES survey that '*Remaining survey will take approx.. half a day; Next workable window: under review.'* From MOC Request dated 16 June 2022, through which KML sought permission to carry out the remaining MBES survey with the Clyde Adventurer, '*its 2-3 hrs to steam to site to for calibration. The calibration requires 30min, then approximately 4 hrs required to perform the survey.*' It

would presumably be necessary thereafter to 'steam' back following the survey i.e. around 10 hours in total. The document made clear that the survey was to be completed at 'High Tide', so this would also affect when the survey could take place.

4) On Wednesday 15 June 2022, the Daily Call notes record '*Next workable window: Thursday/Friday*'. Those in charge had evidently decided that the MBES could not be completed on 15 June 2022 because of weather conditions. Whilst, as set out above, the wave height did not breach the limit on the data agreed by the experts, it is clear that at the time the Daily Call was made (the email following the call was at 9.27am), the wave height was in fact at 0.7 or 0.6m, in breach of the limits, as they had been consistently during the previous week. Perhaps unsurprisingly, those making the relevant decision were not satisfied that there would be a sufficient window to make it safe to embark on the survey until at least the following day.

5) On Thursday 16 June 2022, the Daily Call notes record, '*Next workable window: forecasts to be closely monitored, maybe weekend*.' The decision had evidently been taken that the weather conditions on Thursday, and forecasts for Friday, were insufficiently benign to allow the survey to go ahead safely. It is also clear that the ongoing weather delays potentially required a change in vessel due to availability: as set out in MOC3, "*Due to availability issues the survey is proposed to be completed with alternative vessel Clyde Adventurer.*" That change request needed to be submitted and approved prior to commencing the MBES survey if a different vessel was to be used.

6) On Friday 17 June 2022, the Daily Call notes repeat, '*Next workable window: forecasts to be closely monitored, maybe weekend*.' The same call – that it was not possible to carry out the MBES survey due to the weather conditions – was made on the basis of the information and forecasts on the ground. It is also recorded that, '*MOC for the use of 'Clyde Adventurer' as vessel for MBES Survey to be issued by today (17/06)*'. It appears that the relevant authorisation for change of vessel had not been obtained either, irrespective of the weather.

7) There is no dispute that, on the basis of the weather data, the conditions after 13.00 on 17 June 2022 meant that it was not possible to carry out the MBES survey prior to the point at which it was in fact carried out. This is also reflected in the weekend's Daily Call notes. Moving forward to Monday 20 June 2022, the Daily Call notes record that the ETA for the UTV-670 was midday that day. They also record that the MOC for the use of the *Clyde Adventurer* was with VKL/LOC for review, also due that day.

8) The following day, the Daily Call on Tuesday 21 June 2022 notes that '*All the equipment are arrived*'. In respect of the MBES survey, it notes, '*Next workable window: Thu 23/06*'. In relation to the programme of works, it stated, '*based on later forecasts, opportunity to start the Works on Thu. 23/06'*. Feedback on the outstanding MOC for the use of the *Clyde Adventurer* was due that morning.

9) On Wednesday 22 June 2022, it was recorded in the Daily Call notes that the off-shore mobilisation would be completed that day, with the vessel expected to sail away by 2300 that evening. The next window for the MBES survey was still

recorded as the following day, as was the first opportunity to start the works. Feedback on the MOC evidently had not been given the previous day, as it was recorded as anticipated on 22/6.

10) On Thursday 23 June 2022, it was recorded that the vessel had sailed away, and that on-shore mobilisation was ongoing. The MBES pre-survey was noted as 'on-going', albeit (notwithstanding the MOC) with the LIMBO rather the Clyde Adventurer.

11) On Friday 24, June 2022, it was noted that the MBES Survey was completed '*(awaiting data)*', with Trenching ops Phase 1 expected to commence that afternoon/evening.

105. It follows from this analysis that the completion of the MBES survey was not delayed because of the late arrival of the UTV-670. The survey was stopped on 9 June 2022 due to weather, and not restarted notwithstanding daily reviews of whether the conditions were suitable, until 23 June 2022 by which time the UTV-670 had arrived and been mobilised onto the Suzanne A.

106. This conclusion accords with the evidence of Mr Gamble and Mr Frazer, and it demonstrates that Mr Scullion's assertions to the contrary are wrong. Mr Frazer's recollection was that '*during the time when the Tool was held up by UK Customs, KML was not yet in a position to carry out the MBES survey, because the weather conditions had not permitted it*'. Mr Frazer was taken in cross-examination to the weather data from the DPRs (which is the data the experts considered and agreed did not exceed the limits). He fairly accepted what that DPRs showed as against the weather limits, but when it was put to him on the basis of that data that his recollection could not be right, he reiterated that his understanding remained that it was the weather conditions that were keeping the RHIB out of the water. He was not shown the Daily Call notes, which clearly validate his recollection.

107. Therefore, notwithstanding the delays in customs, the UTV-670 arrived '*in time*'. In all the circumstances, it therefore arrived within a reasonable time, that being a time which did not cause any delay or disruption to the following programme of works which were dependent upon its arrival.

108. In these circumstances, a determination of whether Pharos should bear responsibility for the customs delays is academic. That is because even if Pharos were in breach of its obligations relating to the export of the UTV-670, it did not in fact cause any delay and/or any loss to KML.

109. Notwithstanding this, assuming I am wrong as to the question of a reasonable time and/or causation, and that the absence of the UTV-670 did cause some actual delay, I consider briefly the submissions made by KML in respect of the customs delays.

110. It is not suggested that Pharos or its agents did anything to cause the customs delay, or did not do as much as they could to expedite the delivery once it was clear that there was going to be a delay. The central argument is that Pharos should have anticipated that either there was going to be or at least could be customs delays, and if they had shipped the UTV-670 10 days earlier, any anticipated customs delays could be accommodated.

111. The evidence relied upon by KML is (a) a difficulty encountered by Mr Gamble with customs pulling a shipment of Pharos' equipment in March 2022 in respect of a different project; (b) Mr Walker's reference to other equipment (quite possibly that referred to in (a), given the timing) having been delayed by up to three months in his letter to the Department of International Trade in June 2022; (c) Mr Walker's awareness of potential issues surrounding the export of subsea equipment given his letter '*To whom it may concern*' requesting assistance on 1 June 2022; (d) Pharos' complete, or near complete, lack of previous experience of temporarily exporting the UTV-670; (e) the fact that a standard time for necessary checks by a specialist team in HRMC was stated in communications (whilst the UTV-670 was undergoing those checks) as being 10 days. It is said that had Pharos made appropriate enquiries in advance, it should have anticipated a potential delay of 10 days, which is in fact what the delay was. Had it done so, it should, in order to deliver the UTV-670 within a reasonable time, have shipped the UTV-670 10 days earlier.

112. This argument depends upon a conclusion that it was possible for Pharos to have commenced the shipment process 10 days in advance of 9 June (i.e. 31 May). However, on KML's own case, the date for delivery of 9 June 2022 was only provided to Pharos by a tracker dated 31 May 2022. It is plainly not reasonable to have expected Pharos to arrange for shipment of UTV-670 on the very day it is told the scheduled date for mobilisation. There is no evidence to suggest that Pharos did not, following the agreement of the date 'at around the end of May' (per the evidence of Mr Scullion, pleaded as 31 May 2022 at paragraph 11(2) of the Amended Defence), act reasonably in arranging shipment. Putting this in context, at the date of the provision of the revised Purchase Order on 25 May 2022, the delivery address was still 'TBC'. On the evening of 30 May 2022, Mr Gamble informed KML that '*we will be confirming our order for subsea rental equipment first thing tomorrow, with the intention of taking receipt on 3rd June to allow integration to the vehicle prior to the full system shipment on the 6th June to Esberg*'.

113. Even if, therefore, Pharos should have anticipated the possibility of up to a 10 day delay due to customs issues which it could not control, there is no evidence that Pharos could reasonably have commenced shipment earlier than 6 June 2022 in any event. Given that the events that then followed were out of Pharos' control, and that it did as much as it reasonably could to expedite delivery (indeed – it went beyond what it should have done in its communications), the claims that a reasonable time for delivery was 9 June 2022 and/or that a failure to have delivered by this was caused by a lack of reasonable skill and care on Pharos's part fail.

E. The Embedment Works: The Causes of Downtime

114. A key part of the evidence presented by both experts are their respective downtime analyses. These seek to identify the amount of downtime experienced on the project and what the causes of that downtime were. The amount of downtime is relevant not just to the quantification of any particular issue in the context of causation and loss, but may also shine a light on the extent of a particular problem which itself may be relevant to allegations of breach, and in particular fitness for purpose.

115. Mr Moore's analysis sought to focus on four categories of downtime: weather related downtime, downtime from technical and operations issues associated with Pharos' equipment spread, downtime as a result of tides and downtime as a result of what he called unfavourable seabed conditions which resulted in the UTV-670 becoming trapped on the seabed floor.

116. Mr Scott's analysis instead sought to identify weather downtime and tool downtime, although his figure for this latter period included what he described as 'the Gulley Period' which overlaps with Mr Moore's unfavourable seabed conditions category.

117. The figures obtained by Mr Moore and Mr Scott do not align. This is as a result of methodological differences. It is necessary first, therefore, to consider which experts' analysis I should use when considering issues of, or quantification of, downtime.

Weather Downtime

118. Mr Moore has taken his data from the Pharos Daily Progress Reports. He states that he has also used information available to him from the KML produced DPRs and in the descriptions of the works produced in the Marine Warranty Surveyor report. Fundamentally, however, Mr Moore has sought to identify when the UTV-670 was not, as a matter of fact, operational due to the Susanne-A either being stood down in the port of Hvide Sande or when it had moved to sheltered waters close to the burial site due to bad weather.

119. By contrast, Mr Scott's analysis was based upon a more theoretical approach which compared the weather working limits as set out in the Method Statement with the weather data taken from KML's DPRs. At least from the beginning of July onwards, the weather data recorded within KML's DPRs was forecast data.

120. It is unsurprising that there are some differences, therefore, between the results. Mr Moore's analysis has sought to identify when there was actually downtime due to waiting on weather; Mr Scott's approach sought to identify when a forecast exceeded a weather limit, irrespective of what actually happened. Mr Moore's approach is plainly the more appropriate one for considering when downtime occurred because of waiting on weather. Mr Scott's approach takes no account of the fact that the Master of the Suzanne A would have taken decisions either to stop, or to carry on, work which may not have coincided precisely with a record of what had been forecast to have happened at any particular time. It is what actually happened that is relevant.

121. Mr Moore's analysis attributed 633 hours to weather downtime between the actual commencement of the works, on 23/24 June 2022 to their conclusion on 17 August. He logged 465 hours in July and 167 hours in August. As he describes, the primary contributors to these hours were a period of unfavourable weather across July 2022, which was interspersed with part favourable weather conditions on 13, 14, 18, 19 and 29 to 31 July, where offshore operations continued, and unfavourable conditions from 1 to 8 August 2022.

122. For the reasons I have set out above, the risk for waiting on weather remained with KML. However, plainly that risk only eventuated for the period during which Pharos should reasonably have been carrying out the work. In other words, if (for example) the works ought reasonably to have been completed by the end of July (taking into

account the weather and any other matters appropriate which were not Pharos' risk), then the weather conditions in August 2022 are irrelevant and do not form any basis for a claim for payment from Pharos.

Tidal Downtime

123. The weather downtime analysis by Mr Moore did not include any consideration of waiting on tides (and this is not something considered by Mr Scott at all). Mr Moore identified 39 hours and 18 minutes waiting on tides. The most significant of these was an overall period of 33 hours and 47 minutes during August when work was waiting on tidal conditions particularly so that the Suzanne A could cross the sandbar. In principle, downtime caused by waiting on tides should be taken account of in determining, in accordance with the principles above, what a 'reasonable' time was for Pharos to complete the works. Waiting on tides was out of Pharos' control.

Seabed Condition Downtime

124. Mr Moore's assessment of downtime in respect of what he calls unfavourable seabed conditions amounted to 194 hours 31 minutes. Mr Scott did not separately calculate any period of time which might be associated with seabed conditions, but these periods were included within his general 'Tool Downtime' category. The contents of Mr Moore's report in this respect were not materially challenged, no doubt because KML rely upon its contractual/risk allocation arguments. The following summary of the 5 longest seabed condition related events (accounting for over 177 of the lost hours) in the following paragraphs is drawn largely from his report.

125. **Event 4 - 09:12 on 29/06/2022 to 18:45 on 04/07/2022 (129 hours, 33 minutes)** The UTV-670 became trapped on a high incline of 29° during jetting and burial works at around KP 0.154. from 09:12 on 29/06/2022 and remained so until around 23:05 on 30/06/2022. As a result of worsening weather conditions the UTV-670 could not be driven back to the Susanne-A and had to be parked on the beach while the vessel returned to port. The UTV-670 was then transported back to the port by road arriving at 09:00 on 04/07/2022. The UTV-670 was brought back into operational status at around 18:45 on the same day. The root cause of the downtime including the waiting time at the beach, road transport to port and remediation was recorded by Mr Moore as downtime due to seabed conditions.

126. KP 0.154 is within the very nearshore area of the trenching route. This lies in an area of the cable route between KP 0.14 and KP 0.27 which underwent an MBES survey prior to trenching works starting in the nearshore area. It was KML (or its other sub-contractors) carrying out the MBES survey along the full cable route between KP 0.14 and KP 2.48. There is no dispute that, in fact, as noted in the MBES survey itself the final part of the MBES survey performed on 23 June 2022 was only between KP 0.196 – KP 2.5. The survey boat was not able to survey KP 0.14 – KP 0.195 due to weather restrictions.

127. **Event 13 - 14:08 to 20:05 on 10/08/2022 (5 hours, 57 minutes)** At the start of this event the relevant DPR indicated that trenching work had ceased at KP 0.694, the UTV-670 had been repositioned and was undertaking survey work at KP 0.396 when it became submerged in fluidised soils.

128. According to the DPR records, cable trenching had taken place across KP 0.396 during dive 19 and dive 20, 11 days earlier on 30/07/2022. KP 0.396 lies on the edge of a slope at a seabed depression which falls from 4.7m to 6.0m. Whilst no report of high inclinations was noted in the POG DPR, this seabed feature may have contributed to the TROV becoming stuck.

129. **Event 14 – 11:25-18:52 on 11/08/2022 (7 hours, 27 minutes)** The DPR for 11/08/2022 at 11:29 reports that the UTV-670 was caught in a sink hole at KP 0.309. Because of tidal limitations and the fact that the Susanne-A had insufficient time to transit over the offshore sandbank, the UTV-670 was recovered back to deck. The area adjacent to KP 0.309 on the August 2022 MBES survey results shows evidence of localised depressions, some of which are just outside of the 2m depth offset lines, depicted in dark blue, to the north and the south of the cable route. Mr Moore considered that the depressions may have been the reported sink hole.

130. **Event 19 – 20:16 on 12/08/2022 to 06:20 on 13/08/2022 (10 hours 4 minutes)** From KP 0.261 at 19:54 the UTV-670 started to move along the cable line increasing water pressure on the pumps. At 20:02 (KP 0.277) it is reported that the UTV-670 is trenching steadily at 2.2m depth. Following on from this report around 20:14 (KP 0.296), the DPR indicates that the TROV is out of the ditch and is trenching. The DPR does not report what this ditch was or its severity.

131. At 20:16 (KP 0.296) the UTV-670 is reported as being stuck and an all stop command was issued. The UTV-670 remained stuck until grading in operations were started at 06:20 the following day. This event is close to the KP 0.309 position of Event 4 and therefore this event may have been caused by the same reasons as Event 4. Some 12 minutes after this event the aft of the TROV sank in collapsed soils at KP 0.148 where the vehicle is reported to have experienced roll inclination angles of 22.8° to 24.3°. This second incident (Event 20) incurred further downtime of 1 hour 23 minutes.

132. **Event 21 -14:46 on 13/08/2022 to 14:16 on 14/08/2022 (23 hours 30 minutes)** Shortly after Event 19, the TROV became trapped again at KP 0.182. This was resolved at 15:59 but the vehicle became trapped again in what the POG DPR refers to as 'the same trench' and remained so until 14:16 the following day. The KP 0.182 location is shortly after the position where Event 4 took place to the west. It can be seen from the location of Event 21 the contour lines are close together and change direction significantly, indicating the presence of localised seafloor trenches or depressions. The DPR also reports at 15:42 and 19:25 that (static) roll measurements showed angles of 17.2° on each occasion. This is a further indication of the high seabed slope at this location.

133. I consider in more detail below the question of liability for time spent dealing with unfavourable seabed conditions.

Technical and Operational/Tool Downtime

134. Mr Moore sought to identify all downtime associated with the operation of the UTV-670 and its associated equipment including its breakdown and maintenance. The equipment, for which Pharos was responsible, considered in this analysis was (1) the

UTV-670 including jetting swords, TSS350 survey equipment to measure depth of cable burial (DoB), MBES sonar and GPS antenna; (2) the 8" Oroflex 20 lay flat water hoses and their couplings; (3) the hose spooler unit and section of round hose integral to this equipment; (4) the tether management system (TMS) including umbilical; and (5) the water pumps and suction hoses providing water under pressure to the jetting system.

135. Mr Moore then calculated the total technical and operational related downtime and identified the equipment to which the downtime related as follows:

Causative factor	Downtime (June – August 2022)	
	H	M
TROV Lay flat hoses	36	4
TROV Jetting water supply/pressure	34	16
TROV Sword	11	54
TROV Survey and positioning equipment	10	15
TROV TMS	8	10
TROV Thrusters	3	30
TROV Compensator repairs	2	47
TROV Tracks	2	37
TROV Reconfiguration	1	45
TROV Profiler bracket	1	5
TROV Lamps	1	0
TOTAL		
TOTAL (adjusted to account for mins.)	**113**	**13**

136. Whilst Mr Moore was challenged on some of the views he expressed about responsibility for some of the problems encountered with the equipment identified above, his assessment of downtime to cause was not challenged.

137. Mr Scott's analysis of Tool downtime was contained in Appendix A8 to his report. His evidence about this Appendix was not entirely satisfactory. The Appendix appears to contain a number of semi-rhetorical questions, which Mr Scott indicated were notes that he had made for himself rather than conclusions. Mr Scott was taken to an example of 25 June 2022, where the number of tool downtime hours was identified to be 4. Mr Scott was asked how he had calculated 4 hours, and he answered, frankly, that he had no idea, other than he thought it came from the DPR. It did not, however. The Pharos DPR identifies 2.13 downtime for asset breakdown. Mr Moore's schedule for the same date allocates (by including matters which the Pharos DPR had not specifically identified as asset related downtime, but which Mr Moore considered should be) over 7 hours' downtime, the specific causes clearly particularised. The KML DPR does not identify any specific period of standby or downtime. There is no obvious explanation from the DPRs as to how 4 hours tool downtime might have been arrived at.

138. Mr Scott then candidly admitted that he was only aware that the document forming Appendix 8 had been inserted as an Appendix at the last minute, and, equally honestly, that the Court should not take much notice of it. Mr Scott did commend his conclusions in his Appendix A7 as to downtime, but this Appendix identified hours (e.g. the same 4 hours for 25 June 2022) but contained no explanations as to how the

numbers had been arrived at. It may be for this reason that it was thought important (perhaps by those assisting Mr Scott in the compilation of the report) to include a document which sought to provide at least some explanation of the numbers in Appendix A7. Ultimately, whether or not the figures in Appendix A7 for tool downtime were or were not drawn from Appendix A8, the Court is left in the position that Mr Scott has provided no rationale for his tool downtime analysis, such that it may sensibly be interrogated or accepted. This contrasts with Mr Moore's (unchallenged) analysis which sets out in detail the reasons he sought to attribute certain periods to particular downtime causes.

139. I therefore have no hesitation in concluding that in assessing downtime attributable to causes (which does not include the question of responsibility for those causes) Mr Moore's analysis is to be preferred.

E. KML's Allegations of Breach

140. The pleaded claim in respect of Clause 4 of the warranties set out at paragraph 37(4) of the Amended Defence and Counterclaim alleges that the UTV-670 was not fit for its intended purpose in that:

(1) Throughout the works the UTV-670 suffered various issues including (i) sensor issues; (ii) repeated failure to maintain pressure; (iii) inability to maintain traction; (iv) hose failures; and (v) HPU failures.

(2) On average for every 1.6 hours that the UTV-670 worked 1 hour of breakdown/maintenance time was incurred.

(3) The burial depth of the cable was not achieved throughout the length of the cable (although this appears to relate to a claim for potential future remedial work which has not been pursued).

(4) The tether length installed onto the UTV-670 was 250m rather than 350m (this does not appear to have been pursued).

141. In its pleaded answers to Requests for Further Information, KML set out other equipment which the equipment supplied continually failed ('by way of examples'). Reference was then made to just 3 events, on 26, 27 and 29 June relating to the anchor wires disconnecting: a hose connection failing and a hose fitting parting.

142. Further criticisms of the equipment were made indirectly in paragraphs 37(5) of the Amended Defence and Counterclaim, which introduced an alleged failure on the part Pharos to exercise reasonable skill and care in:

(1) failing to conclude that the UTV-670 was unsuitable for the conditions which it was likely to encounter on site, to have advised accordingly and to have provided alternative apparatus which was suitable for operation in the conditions;

(2) failing properly to assess the suitability of the lay-flat hoses which were prone to sink when filled with water, to strain the rigging securing the buoys used to keep them afloat, and to kink and become entangled with the guideline and the communication umbilical leading to the UTV-670. It is said that Pharos should have deployed appropriate floating hoses;

(3) failing to advise on or before 24 May 2022, that the UTV-670 should be deployed first to work in the offshore area, concurrent with the MBES survey, and to amend the Method Statement accordingly. This would have, it is alleged, minimised the reasonably foreseeable risk of delays and difficulties in the event that the MBES survey of the near and very near inshore areas revealed seabed conditions which required dredging or other remedial works before the UTV-670 would be suitable to operate in those areas.

143. Against KML's pleaded case, I consider the breaches pursued by KML in Closing.

(1) Lay Flat Hoses

144. The water used by the UTV-670's jetting sword was fed into it via an 8-inch water hose which was connected to two water pumps on board the vessel. This was a "lay flat" hose, meaning that it folded down when there was no water in it, similar, as Mr Gamble put it, to an extra-large fireman's hose. The hose used on this project came in 15m sections and needed to be connected via couplings.

145. There is no dispute that the couplings initially procured by Pharos to connect lengths of the lay flat hoses did not work. Although in opening a case relating to these couplings was advanced based on a combination of the urgency of operations combined with the 'ultimate' cause being the requirement for 200m of hoses attributable to the choice of the Suzanne A, this was effectively abandoned when Mr Gamble fairly conceded in cross-examination (as, indeed, he had already accepted at paragraph 67 of his witness statement) that irrespective of these issues, the procurement of sub-standard couplings was an error or failing on the part of Pharos.

146. Mr Moore attributed 36 hours and 4 minutes to 'Lay Flat Hoses'. The majority of the relevant time related explicitly to the coupling issue. The only other entries attributable to the lay flat hoses by Mr Moore were:

(1) 30 minutes on 24 June 2022 ('Completed integration of 30m 8" water hose to vehicle');

(2) 2 hours and 39 minutes on 25 June 2022 ('Drone shows water hose section snagged with the y anchor of the rigging…..Shore team working to release hoses'). This is the same entanglement issue identified by Mr Scott at paragraphs 7.58 and 7.60 of his Report; and

(3) 2 hours 35 on 29 July 2022, after the new couplings had been deployed, ('8" flat lay hose parted at the vehicle…repair maintenance underway').

147. In addition to the specific coupling issue, KML contends that Pharos was in breach by reason of the decision to use lay flat hoses itself, rather than flexible hoses (i.e. like large garden hoses) to supply water to the UTV-670 during very nearshore trenching operations. Although its allegations about the use of lay flat hoses *per se* are pleaded with somewhat greater particularly under its claim for a failure to have assessed their suitability with reasonable skill and care, the issue in dispute is clear, and no point was taken by Pharos as to the pleadings in relation to the criticism made of the decision to use lay flat hoses at all.

148. The reason the distance between the Suzanne A (on which the High Pressure Unit ('HPU') feeding the water to the UTV-670 sat) needed to be up to 200m when the UTV-670 was near shore was the inability of the Suzanne A to get closer because of its draught. Mr Gamble accepted, rightly, that this was known at the time Pharos entered into its contract with KML. Mr Macey-Dare KC is therefore correct that, in these circumstances, whilst it might be considered that the choice of vessel was itself potentially sub-optimal, as considered by Mr Moore, this is irrelevant to the analysis of fitness for purpose or whether Pharos' choice of system was reasonably competent. Pharos undertook to provide a system which worked on the Suzanne A; if it could not do so, then it should not have entered the contract.

149. The allegation of lack of fitness for purpose of the lay-flat hoses in principle, leaving aside the coupling failures, rests on the fact that, unlike flexible hoses which could be kept from sinking with a pillow float system, the flat hoses necessitated a 'taut wire' or 'guiding wire' system for controlling the hose and the umbilical and ensuring it did not sink.

150. Mr Moore agreed that the taut wire was 'sub-optimal'. Mr Moore also agreed that the 200m distance would have presented no significant issues for this operation with the use of flexible hoses. He contended, however, in his report that the use of a lay-flat hose was necessitated by space constraints on the vessel. Notwithstanding this, Mr Moore accepted in cross-examination that it is likely that space would not have been an issue if Pharos had required any unnecessary equipment on the deck of the Suzanne A to be removed, which they would have been entitled to insist upon. I do not therefore consider space to be a justification for the use of lay flat hoses. I note, in addition, that Mr Gamble did not consider that space was the reason behind the choice of a lay flat hose. He said it was because it was lighter when being pulled by the UTV-670 in the water. It is not obvious why this should be the case (with flexible hose supported by floats) and no further detail was given. Weight was not a consideration or justification considered by Mr Moore.

151. In his report, Mr Scott considered bluntly that '*the configuration does not work*'. He also concluded that the configuration resulted in most if not all the handling problems with the tool at the beach and nearshore, as set out in his Appendix 8. Notwithstanding the general unreliability of Appendix 8 for the reasons described above, I have nevertheless reviewed the document to discern the nature and extent, and impact, of the concerns Mr Scott is referring to. It is very difficult, reading Appendix 8 as generously as possible, to identify what Mr Scott is referring to save for two entries on 13 July and 10 August which appear to relate to the hose/guide wire configuration causing problems, and in respect of which he records no downtime against the latter.

152. The mere fact that the solution may, particularly compared with another solution, be 'sub-optimal' does not render the solution of itself unfit for purpose. I do not accept Mr Scott's evidence that '*the configuration does not work*'. To the contrary, there is no reliable evidence that the system did not generally work, once the coupling issue was fixed. The entanglement issue identified by both experts appears, on the basis of Mr Moore's analysis which I have accepted, to have been isolated and resulted in less than 3 hour's downtime. In the context of the project as a whole, this is not an issue that demonstrates that the system was unfit for purpose, particularly as it happened near the beginning of the works, and did not happen again. The only other specific

downtime attributable to lay flat hose related issues, coupling aside, is minimal. Whilst more complicated than using flexible hoses, there is no evidence that the system was necessarily slower: even with lay flat hoses and the taut wire system, speeds in excess of 250m/hour for trenching offshore were achieved (as calculated by Mr Moore in his report). In these circumstances, it is not possible to conclude that the system was not fit for purpose, or that its choice was the result of a failure to take reasonable skill and care.

(2) Supply of the UTV-670 unable to cope with seabed gradients

153. At various times, the UTV-670 was unable to trench given the gradients that were in fact encountered in the very nearshore sandbank, and the soil conditions which it encountered in other locations including the offshore sandbank. It is noted that, without prejudice to its strict entitlement, KML's pleaded position is to accept 14.72 hours for 29 June and 24 hours on 30 June (as part of 4 days) for weather and 'slope inclination'.

154. There is no dispute that, as the parties were aware, the UTV-670 had a 15 degree working limit. KML relies, in addition, on Mr Scullion's evidence, unchallenged, that Mr Taylor and Mr Shaw of Pharos confirmed to him orally that the UTV-670 could work with gradients up to 20 degrees, and upon Mr Gamble's description of the UTV-670 as 'exceptionally manoeuvrable'. KML contends that the delays caused by manoeuvrability issues caused by slopes in excess of 15 degrees, or otherwise in difficult soils, are culpable delays for which Pharos is liable.

155. Mr Moore's analysis demonstrates that just over 194 hours were lost due to soil conditions which caused the UTV-670 to become stuck. He includes these hours within his assessment of a 'reasonable time' for completing the trenching works, and therefore does not consider that this was the result of an inherent unfitness for purpose of the equipment or that way in which the Pharos dealt with the seabed conditions was unreasonable.

156. In relation to KML's principal argument, I do not consider that the fact that the UTV-670 was unable to deal with certain seabed conditions itself meant that the UTV-670 was unfit for purpose or the result of an assessment of suitability which was undertaken without reasonable skill and care. This is principally because such a case is not supported by Mr Scott, KML's own expert, who expresses no such opinion about the UTV-670 (notwithstanding lots of other criticisms of the associated equipment). Similarly, the fact that mobilised sand waves might have made certain areas untrenchable without some localised dredging, as happened ultimately to enable trenching in the very nearshore area, does not mean its choice for this project was negligent (again, an allegation which Mr Scott does not advance). Neither KML nor its expert identify a different piece of machinery or methodology which ought to have been used instead. The UTV-670 demonstrated that it was capable of achieving trenching speeds in the region of those indicated by Pharos: Mr Moore calculates an average trenching speed of 4.98 m/min, or 298m/hour over a distance of 1824m in July 2022, and Pharos had made clear that the actual speed would be dependent upon the in situ conditions encountered.

157. In respect of the areas of gradient over 22 degrees, Mr Scott says, instead, that the UTV-670 getting stuck was entirely preventable, and that he would have expected

Pharos to have reviewed the information provided in respect of what he calls 'the gulley', raised the issue with KML and stopped the tool from trenching in the area before it got stuck. In oral evidence, he repeated (a number of times) that once the troublesome area had been identified or encountered, Pharos should have swiftly moved from nearshore to offshore areas, progressing the work effectively, whilst mitigation was undertaken in the nearshore area before returning. In other words, Mr Scott's focus in respect of seabed conditions downtime issue is not to blame the UTV-670 itself (although he is critical of a good deal of the ancillary kit, which I consider further below) or its selection, but to contend that the Pharos operatives failed in the way they dealt with the issues as they arose. I return to this issue, therefore, further below.

158. KML also relies upon Clause 18 of the Standard Terms and Conditions to dismiss Pharos' reliance upon seabed conditions to justify the time taken to trench certain areas. Although not worded particularly elegantly, the clause stipulates that the Seller, Pharos, shall be deemed to have satisfied itself as to all matters which could affect the progress or performance of its obligations.

159. In the context of an obligation with a fixed time, Clause 18 may be effective to preclude a Seller from seeking a variation to extend time (or from claiming that the time obligation had become at large, given the absence of any effective extension of time machinery, save for limited Force Majeure) in circumstances where they come across a condition which they had not foreseen. However, Pharos' obligation was to complete the works reasonably in accordance with the 'indicative' programme, and this deeming provision clause does not change or narrow this obligation. The speed of the UTV-670 necessarily depends upon the nature of the conditions as they are found to be, and this was at least one of the reasons that the programme was agreed to be '*indicative*'. Deeming a contractor knowledgeable of what the conditions were to be encountered does not mean that those conditions are irrelevant when determining whether the contractor has reasonably progressed the works in accordance with an 'indicative' programme. However, neither is deeming such knowledge to Pharos irrelevant. If, for example, delay is caused not by the conditions themselves but (as Mr Scott effectively says) an unpreparedness for those conditions, then such delay may not be reasonable, and therefore in breach of the obligation to carry out the works reasonably in accordance with the indicative programme.

160. Returning, therefore to Mr Scott's central complaint relating to the way in which Pharos dealt with the nearshore seabed conditions, rather than the UTV-670 itself, this allegation was effectively encompassed in the argument that Pharos failed to advise KML on or around 23 June 2022 to reschedule the trenching works so as to commence work in the offshore section. This contention has a narrower focus than the pleaded case, which is set out at paragraph 37(5)(b) of the Amended Defence and Counterclaim. At (v) and (vi), KML plead that Pharos ought to have advised on or before 24 May 2022, and at all times thereafter, that the Tool should be deployed first to work in the offshore area, and that the MBES survey of the near and very near inshore areas should be conducted simultaneously with, or as soon as practicable after, the commencement of that work; and it ought to have advised KML to amend the Method Statement accordingly. It is said that this would have minimised the reasonably foreseeable risk, which was realised, of delays and difficulties in the event

of seabed conditions which required dredging or other remedial works before the UTV-670 would be suitable to operate in those areas.

161. In light of the pleaded allegation, Mr Gamble's written evidence dealt with the extent to which, if at all, Pharos had advised KML to change the order in which the works should be carried out at any time (although, given the pleading to which he was responding, he was, justifiably, not focussed specifically on 23 June 2022, the case as closed). Mr Gamble said:

> "*I recall discussing the idea of changing the sequence of the works with Jimmy Down (KML's marine supervisor). We were told that this was not possible —- my understanding was that the Marine Warranty Surveyor required the MBES survey to be completed before it would issue a license for us to operate the Tool anywhere (even further offshore where the MBES survey had already been carried out) because that is what the method statement said. The Marine Warranty Surveyor was ABL, an independent third party who was responsible for assessing the risks associated with the project (e.g. risks of property damage, and health and safety risks) and certifying that these risks were at an acceptable level. Compliance with the instructions of the Marine Warranty Surveyor is usually a requirement of the insurers.*
>
> *Because the method statement required the MBES survey to be completed in its entirety before works commenced (and because the Marine Warranty Surveyor would not issue a licence until this was done), KML did not permit Pharos to change the sequence of the works and start trenching from the offshore area, even though the MBES data for this area was already available and the Tool was on site and ready to go."*

162. No evidence was served from Mr Down. Mr Gamble was cross-examined about this evidence. It was put to him that the discussion with Mr Down did not happen. Mr Gamble was clear that it did, and that if Mr Down had wished to change the sequence following the suggestion, he could have done. I accept Mr Gamble's recollection on this point. I regarded Mr Gamble as an honest witness. I note that, contrary to KML's case, Mr Gamble was demonstrably correct in relation to the delays to the MBES survey being caused by weather rather than the absence of the UTV-670, and this gives me comfort that he is, likewise, correct in this section of his evidence notwithstanding the lack of any emails at the time. It is entirely credible that, as Mr Gamble explained, the discussion took place on deck, and that when Mr Down dismissed the idea, Mr Gamble did not consider it appropriate to take the point further.

163. Whilst there is no suggestion that Mr Gamble, or anyone else, raised the point about changing the method statement in respect of the sequence again specifically on 23 June 2022, I do not regard it as unreasonable for Pharos not to have done so, given that the issue had been raised once shortly before and not taken forward by Mr Down. The claim that Pharos failed to take reasonable skill and care in failing to advise a sequence change to mitigate the effect of the seabed condition difficulties therefore fails.

164. Nevertheless, as I have indicated above, a 'reasonable time' must take into account not just the actual circumstances as they were encountered, but (a) the 'indicative' programme of 10 days and (b) the required preparedness for coping with worse than anticipated seabed conditions pursuant to Clause 18. Pharos did not have a 'blank cheque' in respect of its time obligation in the context of dealing with seabed conditions. These factors are taken account of when I consider what a reasonable time for completing the works was further below.

The associated equipment, deck layout and personnel

165. KML contend that the condition of various items of equipment ancillary to the UTV-670 placed Pharos in breach of warranty. The pleading identifies (i) sensor issues; (ii) repeated failure to maintain pressure; (iii) inability to maintain traction; (iv) hose failures; and (v) HPU failures.

166. It is clear on the evidence that the amount of downtime caused by technical failures were caused by the sub-standard quality of some of the ancillary equipment, identified in particular by Mr Scott in his report. Mr Moore identified 113 hours downtime in respect of 166 hours of trenching work, with categories of failure including each of the above complaints. Some downtime due to maintenance and repair would be anticipated. According to Pharos' own information, each system (and the UTV-670 with its associated equipment would be a 'system') would require 4 hours out of every 24 hours up to a maximum of 24 hours. Mr Moore's analysis showed that the UTV-670 carried out 166 hours of actual trenching time. Assuming these were continuous (in reality they were not) this would suggest somewhere around the maximum 24 hours' downtime might be anticipated. Against this, 113 hours' downtime – around 5 times higher - is plainly excessive and substantiates KML's case, supported by Mr Scott, that parts of the ancillary kit were not in accordance with the contractually warranted standard. Whilst Mr Frazer accepted that the number of technical failures experienced were '*probably a little higher than average*', I consider this materially understates the position. I also reject Mr Moore's conclusion- largely based upon what he considers was a reasonable response to the various operational failures encountered by Pharos - that the 113 hours should form part of what might be considered 'reasonable'. Mr Moore's conclusion misses the point: even if Pharos' team responded reasonably, the failures should not have been happening in the first place, and were caused by sub-standard ancillary equipment.

167. I therefore consider that the excessive downtime, irrespective of the manner in which the various failures were attended to by Pharos' personnel, to be Pharos' responsibility. I also accept that some of the sub-standard aspects of the UTV-670 equipment (and, in particular, the TMS and the Hose Spooler) would have led, as Mr Moore accepted in principle, to a general degradation in the progress of the Embedment Works: in other words, they slowed operations down even if they did not cause specifically attributable downtime. This degradation must be accounted for in any assessment of what a reasonable time for the works was.

168. One other area of particular criticism from Mr Scott was the deck layout. This was not pleaded, and, even if sub-optimal, no case was advanced as to how the deck layout of itself caused downtime or slowed operations. Nevertheless, my assessment of a reasonable time for completing the works is based upon an assumed competent deck layout, as suggested by Mr Scott in his evidence.

169. Although Mr Scott supported a pleaded case that Pharos' staff were insufficiently qualified, this allegation was not explored to any great degree (either in the context of breach or causation). To the extent necessary, I prefer the evidence of Mr Frazer, Mr Gamble and supported by Mr Moore that the manner in which the various difficulties were responded to was reasonable. That said, my assessment of a reasonable time obviously assumes that the works are being carried out by reasonably competent and properly qualified personnel.

170. Finally, in light of the submissions made by Mr Macey-Dare KC, I add that I was not particularly assisted by KML's allegations around Pharos's disclosure. KML did not pursue any applications for specific disclosure and there is no suggestion that Pharos's disclosure was in breach of any Court Order. It is not appropriate to draw any particular inference against Pharos. Even if it were, however, it would not add to the analysis above in which I have concluded that the system was, in certain respects, sub-standard in any event.

F. What was a reasonable time for carrying out the Embedment Works?

171. Pharos' claim for payment under the Contract and KML's claim for damages both have as their foundation an analysis of what a reasonable time for carrying out the Embedment Works would be, taking account of the relevant contractual allocation of risk.

172. Neither side has presented any sort of critical path analysis. Prior to closings, I asked both sides to consider specifically what the Court's approach should be to the determination of the correct counterfactual for the purposes of determining how long the works reasonably would have taken if neither of their primary cases were correct (e.g. 10 days fixed, with no waiting on weather, on KML's case, or, as long as it in fact took, on Pharos's case).

173. In Closing, Mr Macey-Dare KC presented four counterfactual analyses by way of a spreadsheet, each of which took different conclusions on key issues as their basis. Two counterfactuals were based upon the UTV-670 arriving, as KML's case had been, on 9 June 2022, one with trenching commencing offshore and the other with trenching commencing nearshore; and the other two counterfactuals were based upon the UTV-670 arriving on 20 June 2022, again with the two variables being nearshore commencement and offshore commencement.

174. Mr Woolgar objected to these counterfactuals, on the basis that they should have been presented sooner and available for interrogation by the experts. However, I do not accept that the submission as to what counterfactual might be adopted given particular findings was a matter for either side's expert opinion. The spreadsheet was a helpful submission on the basis of the factual evidence, at least as contended by KML, as to what the Court can or should conclude would have happened in certain scenarios. It was provided in advance of oral closing submissions and could have been the subject of criticism or observations by Mr Woolgar had he wished. Mr Macey-Dare KC sensibly chose two key variables (the arrival date of the UTV-670 and whether works should have commenced nearshore or offshore) which were the subject of exploration in the evidence and which are matters upon which the Court has reached conclusions.

175. I have therefore adopted Mr Macey-Dare KC's counterfactual spreadsheet approach as a sensible starting point for analysing what a reasonable time was for completing the works and/or causation. Of the four counterfactuals, the last (arriving 20th June 2022 and commencing nearshore) accords with my conclusions as to the appropriate assumptions to include within an assessment of a reasonable time. I shall refer to this as 'the KML Counterfactual', which is set out together with an adjusted version ('the Adjusted Counterfactual') at Appendix A to this judgment. The KML Counterfactual concludes that the works ought to have demobilised on 5 August 2022 rather than 17 August 2022, the actual date of demobilisation, and, as such, that there were 12 days of culpable delay caused by Pharos's downtime and/or slower than reasonable progress as a result of the breaches it alleges.

176. The KML Counterfactual allows, around the periods of bad weather, for 12 days' trenching. It also includes, in addition to the trenching period, two days for dredging of the nearshore sandwave during nearshore operations. Given actual trenching time (in the actual seabed conditions) was 166 hours (6.9 days), I consider this, together with the 2 days for dredging the nearshore sandwaves, a reasonable time, taking account of the original indicative programme, the contractual risk allocation and including a 24 day aggregate period for normal downtime.

177. The Adjusted Counterfactual amends the KML Counterfactual to account for periods of waiting on weather which, for the reasons I have set out below, it underestimates. In making the adjustments I have relied upon the spreadsheet produced by Mr Woolgar in closing ('the DPR Spreadsheet'), which sought to produce a detailed day-by-day analysis of downtime split into the categories as identified by Mr Moore, and which I have broadly accepted for the reasons set out above. The adjustments I have made are as follows:

    (1) The Counterfactual has 12 July 2022 as a working day whereas according to the DPR spreadsheet it is 19:39 hours' waiting on weather.

    (2) The Counterfactual has 13 July 2022 as a working day whereas according to the DPR spreadsheet it is 15:22 hours' waiting on weather.

    (3) The Counterfactual has 21 July 2022 as a working day whereas according to the DPR spreadsheet it is 14:36 hours' waiting on weather.

    (4) The Counterfactual has 28 July 2022 as a working day whereas according to the DPR spreadsheet it is 21 hours' waiting on weather.

    (5) The Counterfactual has 4 August 2022 as a working day whereas according to the DPR spreadsheet it is 13:45 hours' waiting on weather.

    (6) taking account of the foregoing amounting to about 86 hours, I adjust by adding four waiting on weather days.

178. The Adjusted Counterfactual shows that the works, undertaken reasonably and accounting for weather, would have concluded on 9 August, instead of 17 August.

179. I note that no specific allowance has been made in the KML Counterfactual for waiting on tides. However, it is clear from the DPR Spreadsheet that the vast

majority of time for waiting on tides took place on 9 August 2022. The waiting was required in order to cross the sandbar. Pursuant to the Adjusted Counterfactual, the Embedment Works would have been completed sooner and 9 August would have been the day for demobilisation. It is not necessary to factor this time, therefore, into the Adjusted Counterfactual. Whilst there were, in addition, about 5 hours waiting on tides in the preceding period (mostly on 26 June 2022), and I consider this adequately accounted for in the 12 x 24 hour periods allowed for trenching nearshore and offshore. In these circumstances, no further account needs to be taken of tides in the Adjusted Counterfactual.

180. On the basis of the delta between actual completion on 17 August 2022 and that assessed within the Adjusted Counterfactual, there are therefore 8 days of 'culpable delay'. The 113 hours specifically identified downtime from Mr Moore's analysis accounts for about 4.7 days. 3.3 days therefore represents general 'degradation' of the operation caused by the real, but not specifically quantifiable, impact of the breaches which I have found. This, on the basis of all the evidence I have heard, appears reasonable. Moreover, and irrespective of the allocation to particular identifiable issues, the 8 days' delay is also the result of Pharos's failure, in breach of the implied term, to carry out the works in a reasonable time.

181. One benefit of the Adjusted Counterfactual analysis in the context of claim and counterclaim is that it attributes each period of time either to productive working (or permissible maintenance) or waiting on weather: there is no period of concurrency within the Adjusted Counterfactual whilst operational downtime occurs at the same time as waiting on weather.

182. It follows from the Adjusted Counterfactual that Pharos is entitled to payment pursuant to the Contract, including waiting on weather, assuming arrival of UTV-670 on 20 June, and demobilisation on 9 August 2022.

183. There are 8 days of culpable delay which, subject to questions of loss, are culpable delays for which Pharos is liable to KML for breach of contract.

G. Quantification of Pharos' Claim

184. Pharos's claim is set out at paragraph 31 of the Particulars of Claim as follows:

Invoice number	Date	Amount invoiced	Amount paid	Amount outstanding
POG_SI_02174	05/05/2022	£9,750.00	£9,750.00	£0.00
POG_SI_02175	23/05/2022	£19,050.00	£19,050.00	£0.00
POG_SI_02185	30/06/2022	£249,067.54	£106,248.00	£142,819.54
POG_SI_02189	31/07/2022	£325,790.27	£0.00	£325,790.27
POG_SI_02190	31/07/2022	£22,312.04	£0.00	£22,312.04
POG_SI_02194	26/08/2022	£273,780.00	£0.00	£273,780.00
POG_SI_02200	31/08/2022	£18,721.94	£0.00	£18,721.94
		£918,471.79	£135,048.00	£783,423.79

185. The amounts invoiced and outstanding are inclusive of VAT.

186. As can be seen, there is no dispute about the first 2 invoices. A partial payment was made against the invoice dated 30 June 2022, and no payment was made against the remainder of the invoices. Excluding the two invoices in respect of which there is no dispute, the amount invoiced was £889,671.79 inclusive of VAT and £741,393.16 ex VAT.

187. One feature of the manner in which Pharos invoiced, as apparent from the invoices themselves and made clear in the presentation within the Master Spreadsheet, is that there were a number of areas in which Pharos did not reflect the Revised Purchase Order in that:

(1) It did not claim a lump sum entitlement to £24,000 for project management.

(2) It did not claim the £81,000 sum for the additional equipment, but £33,650.76 (including the 10% uplift).

(3) It claimed personnel and personnel mobilisation on the basis of the actual number of personnel rather than on the basis of a fixed personnel rate.

(4) It claimed personnel and employee travel expenses at cost plus 10%. The travel expenses appear to include small sums incurred on a daily basis.

188. The Amended Defence denies the value claimed, and asserts that the sum that ought to have been invoiced is £274,742.05 (ex VAT), calculated as follows:

1	Original Contract Amount	See PO 16249	1	Ls	£ 193,000.00	£ 193,000.00	
2	Excusable Delay - WOW / Slope Inclination	Buyer assessment (Rate from PO)	4	Days	£ 8,800.00	£ 35,200.00	
3	Flights _ Mobilisation	POG_SI_02815	1.05	of	£ 11,184.00	£ 11,743.20	PO 5% Markup only not 10% as involved
4	Equipment - Mob / Demob	POG_SI_02815	1.05	of	£ 3,510.00	£ 3,685.50	Lubber Inv + 5%
5	Equipment - Mob / Demob	POG_SI_02815	1.05	of	£ 17,350.00	£ 18,217.50	Lubber Inv + 5%
6	Flights _ Demob	POG_SI_02190	1.05	of	£ 12,281.76	£ 12,895.85	PO 5% Markup only not 10% as involved
7	Flights _ Change / Rotation	POG_SI_02220				£ -	Not reimbursable (Supplier Default)
	Sub Total					£ 274,742.05	

189. As can be seen, the pleaded approach of KML started with the value of the Revised Purchase Order as if it was a Lump Sum for all the project management, equipment and personnel required. The consequence of this is that KML effectively 'admits' an entitlement on Pharos's part to the £24,000 Lump Sum, and the additional equipment costs in the 'not to exceed' sum of £81,000, even though these sums did not form part of the pleaded claim, advanced by reference to the sums invoiced.

190. Where there is no claim for the £24,000 lump sum project management fee, or additional equipment costs at £81,000, I have not, in assessing the amount due, re-instated these sums.

191. It is therefore necessary to consider the following categories of costs claim for the relevant duration (daily costs) and fixed costs:

(1) Equipment spread costs;

(2) Personnel costs (excluding expenses);

(3) Personnel daily expenses;

(4) Transportation of the UTV-670 and other equipment;

(5) Additional equipment.

(6) Transportation (flights) for personnel.

(1) Equipment spread costs;

192. The relevant period for operations and waiting on weather is from 20 June 2022 to 9 August 2022 inclusive. For the avoidance of doubt, this should include the demobilisation day (as appears to be accepted in the Counterfactual calculations, but is the correct approach in any event). This is 51 days. The amount due is **£216,750** (ex VAT).

193. For the avoidance of doubt, this excludes £34,000 claimed under invoice 02194 for 'UTV-670 transit', between 19 August 2022 and 26 August 2022. There is no explanation of the contractual or factual entitlement to this sum by way of pleading, evidence or submission.

(2) Personnel costs (excluding expenses)

194. To the same 51 day period, should be added one day for mobilisation. The amount calculated by reference to Tab 5 of the Master Spreadsheet ('D's position C's Inv (Daily)), column E (Pharos Personnel, total calculated by actual number of personnel pro-rated to £4,550) from 19 June 2022 to 9 August 2022 inclusive is £308,100. This could be said to overstate the entitlement because in reality, it can be seen from the spreadsheet that the number of personnel dropped by 2 during the last 7 days (e.g. towards end of works and for the post burial survey). The costs claimed by reference to the Adjusted Counterfactual should reflect the same reduction (which the actual figures for the period leading up to 9 August 2022 obviously do not reflect). I therefore deduct 2 x 7 x £650 from the figure above (£308,100-£9,100): **£299,000** ex VAT.

195. For the reasons set out at paragraph 79 above, I do not allow the claim for payment for personnel deployment prior to one day before 20 June 2022.

(3) Personnel daily expenses

196. Although there are a number of claims for generally small sums for personnel daily expenses (excluding flights), it is not clear what the contractual basis of the claimed sums is. Pharos have called no evidence in relation to this aspect of its claim, and the submissions do not provide any assistance (either as a matter of principle or quantification).

(4) Transportation of the UTV-670 and other equipment;

197. This was claimed in Invoice 02185 at **£21,903.00** ex VAT. Although the uplift calculated in the invoice is under a column marked '10%', the sum calculated is in

fact 5% as set out in the Revised Purchase Order. This appears to be accepted by KML in the calculation in the calculation in the Amended Defence.

(5) Additional Equipment

198. This was claimed in Invoice 02185 at **£33,650.76** ex VAT, including 10% uplift. This was less than the sum allowed in the Amended Defence.

(6) Transportation (flights) for personnel/employee

199. The sums claimed, ex VAT, for flights within the Pharos invoices appear to be:

(1) Invoice 02185: £11,184.11 plus 10% = £12,302.52 ('personnel');

(2) Invoice 02190: £12,281.76 plus 10% = £13,509.94 ('personnel')

(3) Invoice 02190: £4,841.36 plus 5% = £5,083.43 ('employee')

(4) Invoice 02200: £8,388.86 plus 10% = £9,227.75 ('personnel')

(5) Invoice 02200: £6,070.35 plus 5% = £6,373.87 ('employee')

200. The pleaded position of KML is that it accepts Pharos's entitlement to items (1) and (2) above, subject to an uplift of 5% rather than 10%. By inference, the remaining sums appear to relate to further travel when employees were mobilised and demobilised given the extended duration of the works.

201. At Tab 6 of the Master Spreadsheet provided in opening, a series of points have been taken by KML in relation to the sums admitted in the pleading (e.g. that the location from which it was permissible to mobilise an employee should be restricted to LHR, rather than from elsewhere in the world).

202. It is not appropriate for the Court to determine these submissions insofar as they relate to categories (1) and (2) above, raised in opening for the first time, in contradiction with the pleaded position and no application to amend. This does not extend to the (pleaded) dispute as to the appropriate uplift.

203. It is equally inappropriate for the Court to determine an entitlement to items (3) to (5) in the absence of any evidence or submissions from Pharos as to the factual or legal entitlement to these expenses.

204. In light of the foregoing, the sums allowed are £11,184.11 and £12,281.76, each plus 5%, exclusive of VAT. This amounts to **£24,639.16** ex VAT.

205. The total to which Pharos is due in respect of the disputed invoices, subject to deduction of the amount paid in respect of them and subject to any set off pursuant to the Counterclaim, is:

£216,750.00
£299,000.00

£21,903.00
£33,650.76
£24,639.16
£595,942.76 ex VAT

206. In respect of this, the sum of £106,248 (VAT included) has been paid. The equivalent amount ex VAT is £88,540. The amount owed, subject to set off, is therefore **£507,402.76**.

H. Quantification of KML's Counterclaim

207. The KML Counterclaim as pleaded is calculated on the basis of an average day rate for the vessel and support vessel spread, personnel costs and expenses, multiplied by the number of days' delay, including waiting on weather, claimed to be the responsibility of Pharos. The overall claimed day rate (ex VAT) originally pleaded amounted to £21,820.35. The pleaded claim was calculated, for reasons not explained, on a VAT added rate. On the basis that KML is capable of recovering VAT paid, its losses should be calculated on an ex-VAT basis.

208. The figures originally pleaded were amended pursuant to an application advanced of the first day of trial, and ultimately acceded to, with the new figures set out day by day at Tab 4 of the Master Spreadsheet. These figures were, in Closing, brought forward into the quantum calculation within the Counterfactual. The average spread cost per day claimed by KML taken from the Master Spreadsheet, Tab 4, column H, is £21,217.21. Assuming 8 days' culpable delay caused by Pharos, this amounts to £169,737.68. An extract from the Master Spreadsheet, Tab 4, for 20 June 2022 (by way of example) showing the categories of cost and the ex-VAT GBP converted costs claimed by way of example is set out below:

20/06/2022	SUSANNE A Charter Hire (Standby Rate)	GBP	7,310.00
	SUSANNE A Steward	GBP	301.00
	SUSANNE A Personnel Allowance (non-crew)	GBP	890.10
	SUSANNE A USBL Pole	GBP	86.00
	SUSANNE A Bunkers	GBP	893.81
	SUSANNE A Harbour Fees	DKK	54.14
	CLYDE ADVENTURER Charter (Standby Rate)	GBP	1,464.00
	CLYDE ADVENTURER Bunkers	GBP	334.80
	KML RIB Charter	GBP	260.40
	KML RIB Bunkers	GBP	240.00
	AEQUOREA Positioning Device (Standby Rate)	GBP	1,094.40
	AEQUOREA Positioning Device (Personnel)	GBP	1,680.00
	AEQUOREA Positioning Device (Expenses)	GBP	18.89
	Rotrex - 60T Trailer Winch	GBP	730.00
	KML Personnel (Food)	GBP	133.66
	KML Personnel (Accomodation)	GBP	722.31
	KML Personnel (Travel / Flights)	GBP	893.22
	KML Personnel (Wages)	GBP	3,484.34
20/06/2022		**GBP**	**20,591.07**

209. At the same time as amending the figures within the pleading, KML sought to rely upon a new extract from its SAGE software system, said to show the attribution of various internal expenses and costs to the Viking project, from which the sums claimed for KML Personnel (in the last 4 rows) were claimed.

210. As Mr Woolgar noted in both his written oral closing submissions, and as candidly accepted by Mr Macey-Dare KC in his submissions, KML adduced no witness evidence whatsoever in support of these claimed costs. Mr Scullion's brief evidence in chief relating to the replacement SAGE document did not add substantively to the

quantum related factual witness evidence. Neither side has called a quantum expert to interrogate or audit, whether in support of or to undermine, the claimed costs. KML is therefore relying upon the documents disclosed in support of the costs as proof on their face of the quantum of the counterclaim. Each document upon which reliance is sought to be placed is set out in the Master Spreadsheet, Tab 4.

211. Mr Woolgar submitted that the documents do not speak for themselves. In a schedule produced to accompany his oral closing, he summarised the difficulties each claimed category of cost faced. Mr Woolgar reminded the Court of the well known dicta from Lord Goddard in Bonham Carter v Hyde Park Hotel (1948) 64 TLR 177 in which the Court observed:

> *"Plaintiffs must understand that, if they bring actions for damages it is for them to prove their damage; it is not enough to write down particulars, and, so to speak, throw them at the head of the court, saying: ' This is what I have lost; I ask you to give me these damages.' They have to prove it."*

212. Nevertheless, it is to be remembered that notwithstanding this statement, Lord Goddard in fact found it possible to arrive at a conclusion despite, in that case, the extremely unsatisfactory evidence as to damages, as noted by Edmund Davies LJ in Ashcroft v Curtin [1971] 1 WLR 1731.

213. In relation to the Suzanne A charter hire, steward and personnel allowance (non-crew), the submission boiled down to an acceptance by Mr Woolgar of substantiation of the rates claimed, on the basis of the production of the underlying Supplytime charter, but an argument that in the absence of proof of payment, the claim should fail. I do not accept this. The production of the invoices and underlying agreement are sufficient to establish the relevant liability and, on balance of probabilities, likely loss sustained on a daily basis by delay to the work.

214. It was claimed that the document provided to support the bunker rate was incorrect, but this was, as clarified, a typographical error, and the actual invoice does support the €1,055 day rate.

215. There is a claim for €100 per day for 'USBL Pole'. As Mr Woolgar submits, there is no explanation of what this is or why. In the absence of evidence, this cost fails.

216. In respect of harbour fees, the sums claimed for the preceding period are, as Mr Woolgar submits, peculiarly variable and there is no explanation of how they have been calculated. Perhaps more importantly, no costs are claimed for the relevant period (9 August 2022 onwards). It is not clear, therefore, that additional harbour fees were incurred in the period of culpable delay. These sums should not be part of the daily average cost for the relevant period of delay.

217. In respect of the Clyde Adventurer, the document relied upon for the period 10 June to 22 June standby and operational rates is just a Purchase Order. Whilst this in itself may be insufficient, it is clear that the respective daily rates generally remain the same in the later periods for which there are invoices evidencing KML's liability. Although Mr Woolgar submits there is no substantiation of the application of the discounted rate claimed for a period, the invoice relied upon clearly identifies, by way of note in red, the charter rate discount given. Moreover, this is a reduction from the rate

otherwise payable and therefore works in favour of reducing the costs payable (and the overall average rate obtained). The Clyde Adventurer costs are satisfactorily proved, save in respect of crew change and accommodation. I accept Mr Woolgar's submission that there is no evidence explaining how or why a liability arose for the Clyde Adventurer accommodation, or Travel for Crew Change which appear from 1 July onwards or the basis upon which this was agreed. I also accept that the documentary and factual substantiation for a KML Rib Charter throughout the period is unsatisfactory.

218. There is no explanation for the application of £730 a day for the entire period for the hire of the 60-tonne trailer winch on the basis of a single invoice which is dated 17 April 2022, i.e. before the relevant works. This should not be included.

219. As for the Aequorea positioning device rates now claimed, the Amended Defence was pleaded on the basis of 50% of the rates claimed, not the full rates, and there is no evidence or submission as to why the originally claimed, reduced amount (accompanied by a statement of truth) is incorrect. The rates are substantiated, but the application of the rate should be as per the pleading, at 50%. This should also apply to the Aequorea Down Manned Rate (which did not feature at all in the original pleading), in the absence of any explanation as to why the same approach would not be correct. The only point taken for Aequorea personnel is proof of payment for the August 2022, and for the reasons I have set out above, reliance upon the invoice is sufficient to establish liability and, on balance of probabilities, loss. There is no explanation for the introduction of claimed Aequorea expenses, which did not form part of the original pleaded average rate, and these are to be excluded.

220. The combined £5,233.53 day rate for KML personnel (food, accommodation, travel/flights and wages) relies upon the newly produced SAGE document. This document simply provides a long list of entries of costs, and a daily average cost calculated for each category: £133.66 for food, £722.31 for accommodation, £893.22 for flights and £3,484.34 for wages.

221. The food category is simply a list of entries. There is no basis upon which to validate the connection between these entries and the relevant project, or relevantly, any particular time period within the overall project. The same applies to accommodation. KML have called no witness to support the claims and has not been possible for Pharos (or the Court) on the scant information provided properly to interrogate the information. The appropriateness of the rate claimed has not been established on the balance of probabilities.

222. As for flights, there is no evidence, or submission focused on the Sage entries, that the 8 day delay caused any specific additional flights to be taken. There are lots of heads of travel related cost which bear no obvious relationship with loss caused by the culpable delay KML have established. KML have not established the reliability of the costs claimed under this head. It is not possible on the basis of this information reliably to interrogate the average obtained and the appropriateness of its applicability to the period of delay.

223. Finally, the 'wages' category contains lots of entries the basis of which is simply not intelligible. The personnel are usually not identified, so there is no way for a third party to interrogate whether the personnel were linked with the project and there is no

evidence, or even submission, which assists in validating this head of cost. It is not possible to arrive at a reliable figure on the basis of the information provided, and KML's expenses claim in this respect fails.

224. When these adjustments are made to each of the days throughout the project, the average recoverable rate is calculated as £14,151.83. This is set out in Appendix B (Revised Master Spreadsheet Tab 4) to this judgment. Applied to a culpable delay of 8 days, the quantum of KML's claim is **£113,214.63**.

I. Conclusion

225. Subject to the question of set off, and interest, Pharos is owed the sum of **£507,402.76** plus VAT on the basis of unpaid invoices. This amounts to **£608,883.31** inclusive of VAT.

226. KML's counterclaim for damages succeeds in the sum of **£113,214.63**, which may be set off against the sums owed pursuant to paragraph 225 above.

227. As to interest, KML have admitted at paragraph 35(3) of its Amended Particulars of Claim, that if, contrary to its position and sums are due and owing to Pharos (as is the case), in principle the Late Payment of Commercial Debts (Interest) Act 1998 applies. The parties are to seek to agree interest and if not agreed, the Court will determine the matter.